DUT Call for Proposals 2023

Full-Proposal: Consortium and General Information[1]

1. Project overview

Project Short Title/Acronym: *COLINE*			
Project Full Title: *Complex Links of Neighbourhoods*			
Project Coordinator: HUN-REN Centre for Economic- and Regional Studies, Balázs Lengyel			
Main R&I approach: ☒ Research-oriented approach (ROA) ☐ Innovation-oriented approach (IOA)			
Main Transition Pathway: ☐ PED Transition Pathway ☒ 15mC Transition Pathway ☐ CUE Transition Pathway			
Call topics: ☐ PED topic 1: Energy Resilience and Energy Poverty ☐ PED topic 2: Urban Regeneration and Refurbishment ☐ PED topic 3: Enabling Systems for Local Energy Transitions: Collaboration and Sustainable Investment ☐ 15mC topic 1: Integrated Policies and Evidence to Reduce Car-dependency ☒ 15mC topic 2: Mobility and Planning Policies for Proximity-oriented Developments ☐ 15mC topic 3: Empower People for Urban Mobility Transitions ☐ CUE topic 1: The Built Environment as Resource Base ☐ CUE topic 2: Knowledge and optimisation of resources flows between urban and rural areas ☐ CUE topic 3: Planning and Designing urban areas with Nature: Towards a Regenerative Urbanism			
Please enter max. 5 keywords describing your project.	Keyword 1: cell-phone tracking Keyword 2: amenity mix in neighbourhoods Keyword 3: modes of transportation Keyword 4: work-from-home Keyword 5: experienced segregation		
Total Project Costs in EUR:	1.563.376,00	Requested funds in EUR:	1.421.320,00
Duration of the Project in months (max. 36):	36	Expected start:	[01.2025]
Total Effort in Person Months:	182,4	Expected end:	[12.2027]

[1] Detailed financial information must also be given in the Financial Information section on *uefiscdi-direct.ro*

2. Abstract

Compact cities require amenity-rich neighbourhoods. To reduce traffic, urban planners increasingly support the development of dense urban areas combining residential units with walking-distance amenities in aesthetically pleasing and safe neighbourhoods. Yet, our understanding of what makes an attractive neighbourhood, and how changes in the amenity composition of neighbourhoods and the mobility of individuals contribute to urban segregation and climate change goals remains limited. The goal of COLINE is to advance a comprehensive and data driven view of cities combining architectural, mobility, and economic considerations. We will map the amenity composition of neighbourhoods by tracing visits using location-based mobile phone data and will combine these traces with urban perception data collected by crowdsourcing and machine learning (AI) methods. This comprehensive view will allow diagnosing the completeness of amenities around home and work, the multimodal accessibility and experienced segregation of neighbourhoods, and their appeal and resilience at a fine spatial resolution. The research will be conducted in collaboration with the cities of Budapest, Copenhagen, Toulouse, Turin, and Vienna and will include the development of datasets and visualizations designed to power public engagement and open data efforts. This project will advance our understanding of how to create citizen oriented multimodal views of cities.

3. Project Consortium

	Organisation	Type of organisation	Country / Funding agency	Contact Person (first name and family name)
Project Coordinator/ Main Applicant	Közgazdaság-és Regionális Tudományi Kutatóközpont	Public Research Organisation	Hungary / NKFIH	Prof. Balázs Lengyel
Project Partner 2	CEU Gmbh.	University	Austria / FFG	Prof. Márton Karsai
Project Partner 3	Toulouse School of Economics	University	France / ANR	Prof. César A. Hidalgo
Project Partner 4	Danmarks Tekniske Universitet	University	Denmark / IFD	Prof. Laura Alessandretti
Project Partner 5	Istituto Per L'Interscambio Scientifico	Private Research Organisation	Italy / MUR	Prof. Riccardo Di Clemente
Project Partner 6	Budapesti Corvinus Egyetem	University	Hungary / NKFIH	Prof. László Lőrincz
Project Partner 7	BKK Budapesti Közlekedési Központ Zártkörűen Működő Részvénytársaság	City Authority	Hungary	Mr. Tamás Halmos
Project Partner 8	Magyar Telekom Nyilvánosan Működő Részvénytársaság	Large Enterprise	Hungary	Mrs. Anita Hodosán-Hartai
Project Partner 9	Budapest Főváros Önkormányzata	Municipality	Hungary	Dóra Anna Kókai
Project Partner 10	Urban Innovation Vienna Gmbh.	City Authority	Austria	Gerald Franz
Project Partner 11	Toulouse Metropole	Municipality	France	Audeline Rauna

Project Partner 12	5T Torino	Large Enterprise	Italy	Giandomenico Gagliardi
Project Partner 13	Copenhagen Commune	Municipality	Denmark	Anna Garrett
Project Partner 14	Toretei	Small Company	Italy	Elizaveta Gazeeva

4. Excellence of the project: quality of work, objectives, and targets

4.1. Project objectives and targets with respect to the state-of-the-art

By reducing mobility, the COVID-19 pandemic highlighted the importance of local neighbourhood services. This exogenous event accelerated a trend towards *New Urbanism* (Duany and Plater-Zyberk, 1992, 1994, Montgomery, 2013), an urban planning movement that promotes the development of compact mixed use walkable neighbourhoods. Together with climate change pressures, this has led to a move towards active forms of mobility and the notion that amenities should be accessible within short distances (Burton, 2000). The "15-minute city" represents a revival of neighbourhoods that aims not only to decrease urban traffic but to also improve the health of cities and local communities (Allam et al., 2022, Moreno et al., 2021). Yet, while many cities are making efforts towards this direction, it is hard to track progress among multiple dimensions.

While important, existing efforts to help quantify progress towards these goals have some important shortcomings. First, much work on the "15-minute city" quantifies access to amenities using pre-defined walking distances, failing to consider visits to actual amenities and mobility flows (e.g., Calafiore et al., 2022). Only a few recent efforts have incorporated multimodal mobility traces of individuals to urban amenities (Zhang et al., 2023). Graells-Garrido and colleagues (2021) use origin-destination matrices to show that certain amenities (like education and retail) attract more visits from other neighbourhoods, whereas Abbiasov et al. (2024) document that only a small portion of daily trips to amenities are within the 15-minute circle, and that proximity matters more in high-density neighbourhoods than in peripheral neighbourhoods. In sum, the amenity portfolio and location of urban neighbourhoods should matter for the effectiveness of the "15-minute city" developments, but the exact relationships between mobility and amenities is not yet fully understood.

Second, amenities and mobility are not only related to one another, but to the aesthetic appeal of neighbourhoods, as captured in studies of urban perception (Dass et al., 2023, Naik et al., 2017, Salesses et al., 2013). For instance, De Nadai et al. (2016) combined mobility data with urban perception estimates for Italian cities to show that women and older people were statistically more likely to avoid unsafe looking places. Still, the links between multimodal mobility, urban perception, and the amenity mix of neighbourhoods have been not fully explored. Both mobility flows and neighbourhoods specialized in non-ubiquitous amenities concentrate in city centres (Juhász et al., 2023, Xu et al., 2020), which are also often attractive from an urban perception perspective (Naik et al. 2017). This correlation presents a challenge for the development of polycentric cities and for peripheral neighbourhoods, which may lack this trifecta of factors. Third, car traffic can be reduced by mixed modes of mobility (Marshall and Banister, 2000); yet how active mobility can be complemented by public transport to increase access to amenities has been overlooked. Analyses should investigate multiple layers of mobility networks that link neighbourhoods of complementary services and social groups. Fourth, the limited range of mobility around home can lead to increasing segregation in terms of socio-economic strata (Yabe et al., 2023a, Abbiasov et al., 2024). Providing accessible, diverse, and attractive neighbourhoods can contribute to fostering social interactions and balance segregation (Mori et al., 2023). Yet, which amenity portfolios may impact the mixing of socio-economic groups and where in the city is unclear. Furthermore, how decreasing traffic and safer roads could improve gender inequalities in active mobility is still unknown (Gauvin et al., 2020; Battiston et al., 2023). Finally, the "15-minute city" revolves around home neighbourhoods, but not much is known about commercial neighbourhoods. Because commuting contributes a large portion of urban mobility and can also create more inclusive cities (Bokányi et al., 2021), a better understanding is needed about the role of amenities around workplaces.

The goal in this project is to understand how the amenity mix of neighbourhoods, multimodal mobility, and neighbourhood aesthetics and safety contribute to better "15-minute cities". COLINE will investigate these problems in Budapest, Copenhagen, Toulouse, Turin, and Vienna. Where available, we will analyse very detailed, diverse, and sometimes sensitive personal data, but will also collect open-access (OA) and commercially accessible data that are applicable elsewhere, to create scenarios for a series of cities.

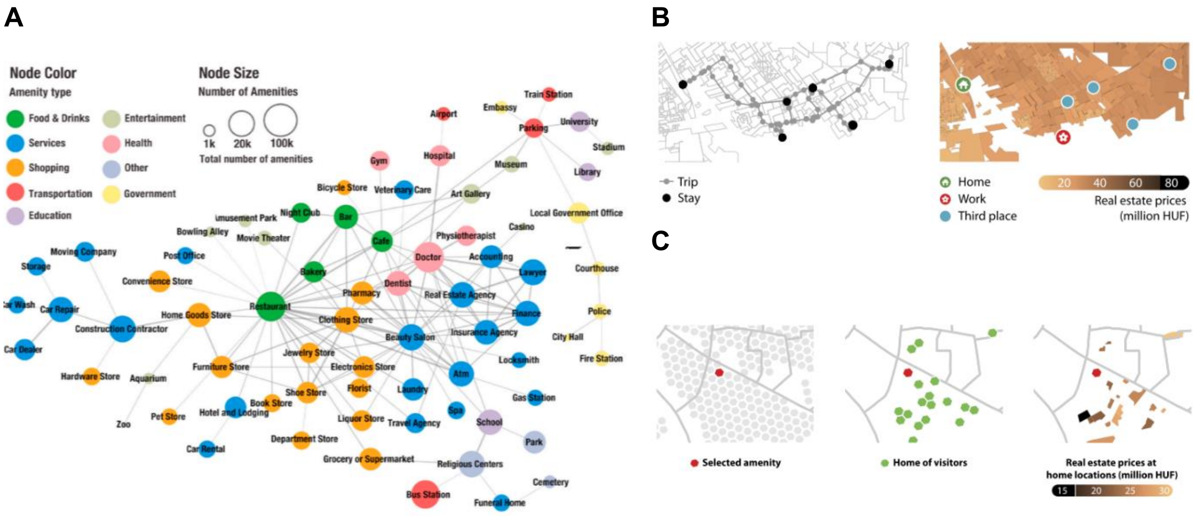
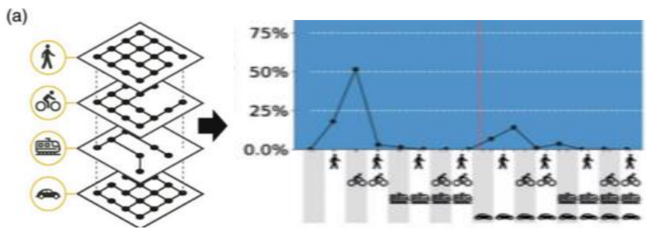


Figure 1. Measuring completeness and socio-economic phenomena of neighbourhoods by amenity visits. **A)** Previous measurement of amenity complementarity was quantified by a network of amenity co-locations, in which amenity types are connected if they are frequently co-located in the same neighbourhood (Hidalgo et al., 2020). **B)** High-frequency mobility traces allow for the identification of home, work, and third places from mobile-phone data (Juhász et al., 2023). **C)** Visitor trips can be linked to amenities that allows for analyses of spatial coverage, travel, and social mixing and for new measurements of amenity complementarity.

The project will advance the state-of-the-art understanding by the following innovative objectives that include methodological developments and analyses of important aspects of the “15-minute city” as well.

Objective I. Neighbourhood characterization by amenity-space based on visits and by perceptions: Previous work on the amenity mix of urban neighbourhoods has used the co-location of services to generate networks of amenities (Figure 1A) for the analysis of neighbourhood development (Hidalgo et al., 2020). Here, we will link high-frequency cell-phone data to amenities to observe individual amenity visits (Figure 1B,C). This will enable us to characterize the complementarity of local services by the sequences of amenity visits (Di Clemente et al., 2018, Yang et al., 2023, Yabe et al., 2023b). Significant efforts will be devoted to choosing and combining data sources that will allow comparisons of amenity spaces across neighbourhoods and across cities. GPS data of smart-phone applications allow for precise geo-location (Juhász et al. 2023) and are available in many European cities (for example from providers SafeGraph, Spectus or Near), while cell information covers much larger proportions of urban dwellers but provides slightly less accurate geo-location (González et al., 2008) and is available in fewer cities. We explicitly address this issue in *Objective V*. The quantification of urban perceptions has been accomplished at scale by Naik et al. (2017) by combining Google Streetview data with generative AI, and online experiments. Since this data is increasingly problematic to work with in scientific research, we will collect street visuals in chosen locations within the investigated cities, modify these images and measure perceptions in participatory workshops to understand which characteristics of urban landscape is received as attractive, for which social strata, and safe for active mobility and for which demographic group (De Nadai et al., 2016). The new amenity space and urban perception measures will be used to characterize neighbourhood completeness and attractivity in detail and to explain why some urban areas are more resilient and flourishing than others.

A



B

Figure 2. Multilayer mobility networks and experienced segregation in cities. **A)** The combination of transportation modes generates accurate accessibility networks ( ). **B)** Social mixing in urban locations can be inferred from home locations and individual mobility (Fan et al., 2023).

Objective II. Modes of transportation to amenities and emissions: Amenity visits will be decomposed by travel time from home and by mode of transport to create new accessibility metrics for the "15-minutes city". First, we will combine traditional routing techniques of active mobility on OpenStreetMap with openly available public transport timetable data (GTFS) to estimate time to reach different amenities from home by multimodal mobility. This will allow us to better characterize neighbourhoods by available amenities and to test how, and where in the city, amenity portfolios change if we allow for increasing time of active mobility following the "x-minute city" concept (Logan et al., 2022). Then, we will measure the speed of travel to amenity visits from cell-phone traces and will estimate the mode of transportation to each amenity visit by leveraging GTFS data from public transport providers and with data of bike-sharing services. Distinguishing car traffic, public transport, and active mobility, we will generate multiple layers of observed mobility networks from these trips (Figure 2A) (Alessandretti et al., 2016, 2017, 2023, Edsberg et al., 2022). Analysing the dynamics of the multilayer mobility networks, we will aim to identify causal relationships between changes in one layer in the network (eg. due to metro reconstruction or opening a new tram line) and changes in the other layer (eg. increasing or decreasing car traffic). We will quantify whether such changes in dependent layers are more pronounced across neighbourhoods that have complementary or similar amenity profiles. Finally, we aim to identify missing links in the public transportation networks that can maximize the reduction of car traffic to amenities (Allam et al., 2022b). Using parameters from previous studies on the type of polluters and the level of pollution in specific roads (Böhm et al., 2022) and consulting with public transport companies and municipalities, we will simulate alternative scenarios to provide estimates about how changes in public transport can change the carbon emissions at the city level.

Objective III. Experienced segregation and inclusion: Because limited range of mobility can increase socio-economic segregation, harmful especially for low-median households (Abbiasov et al., 2022; Yabe et al., 2023a), we will explore how enhancing amenity mix and accessibility by public transport affects socio-economic integration. We will focus on the placement of amenities that blend different socio-economic groups (Athey et al., 2021; Chetty et al., 2022) and utilize multimodal data including census, real estate, satellite, and street-view data to characterize households and assess social mixing at these amenities (Abitbol and Karsai 2020; Espín-Noboa et al., 2023). Additionally, we will modify the visuals of these areas using generative AI tools and hold participatory workshops in target cities to evaluate how new investments might alter location perceptions and enhance their appeal to diverse socio-demographic groups. We aim to examine the statistical links between accessibility through active mobility, public transport, and the potential for increasing socio-economic mixing. Addressing demographic inclusion in active mobility is vital for the "15-minute city" concept, yet balancing gender inequalities remains challenging. Reports suggest women are less inclined towards active mobility in unsafe conditions (Battiston et al., 2023; Yuan et al. 2023) and exhibit more concentrated mobility patterns than men (Gauvin et al., 2020; Macedo et al., 2022). We will use generative AI to assess environments women deem safe for active mobility, analyse the spatial distribution and accessibility of these areas, and identify necessary modifications to boost their attractiveness for active mobility across different gender and age groups.

Objective IV. The role of workplaces and work-from-home: Citizens spend most of their active time in their workplace but in some cases are also able to work from home. Commuting to work increases pollution since the majority of commuting to work is done by car or by public transport in most European cities (Schwanen, 2002) but commuting facilitates social mixing (Bokányi et al., 2021). Cities are complex economies (Di Clemente et al. 2021, Jara-Figueroa et al., 2018) where the spatial mixture at workplaces is key to generate social interactions that can lead to innovation and economic growth (Abbasiharofteh et al. 2023, Alessandretti et al., 2018, Eriksson and Lengyel 2019, Sapienza et al. 2023). Yet, we still need to better understand how the place of work impacts the "15-minute city" concept. To accomplish this task, we will identify the work location from GPS and cell-information, besides identifying home, and will compare amenity visits around home and work. Then, we will use administrative data on firms and their employees to characterize the work-from-home potential of work locations and will use census data to characterize home locations with the same measure and by most frequent modes of mobility to work. Finally, we will calculate the time spent in home neighbourhoods as a function of amenity complementarity between neighbourhoods of home and work, distance between home and work, typical mode of transport for commuting, and work-from-home capacity.

Objective V. Inter-city learning and Open-Access inference across European cities: Cell-information of phones, and administrative data about firms and employees are restricted to few investigated cities only (Table 1). Thus, we will explore the predictive power of OA and commercially available data in the presence of these additional details and search for correlations between measurements that we make with sensitive or unique data and those that are done with open-access data or smaller scale mobility data and street-view data available elsewhere too. Assuming similar distributions of detailed restricted information across cities, we will input randomized values and test OA correlations between investigated cities. To create insights for further European cities, we will search for additional data sources in a series of cities to replicate findings on amenity complementarity, multilayer transportation networks, experienced segregation, and the role of workplaces.

Table 1. Data sources to be used in investigated cities include restricted (R), commercially available (CA) and open access (OA) data.

Data sources	Cities involved	Objectives
GPS mobility data (CA)	Budapest, Copenhagen, Toulouse, Turin, Vienna	I, II, III, IV, V
Amenities and OpenStreetMap (OA)	Budapest, Copenhagen, Toulouse, Turin, Vienna	I, II, III, IV, V
Cell-information mobility data (R)	Budapest, Vienna	I, V
Street images (OA)	Budapest, Copenhagen, Toulouse,Turin, Vienna	I, III, V
GTFS and bike sharing (OA)	Budapest, Copenhagen, Toulouse, Turin, Vienna	II, III, V
Administrative firm and employee data (R)	Budapest, Copenhagen	IV
Census (R)	Budapest, Copenhagen, Toulouse, Turin, Vienna	IV

We will collect and manage data sources to cover periods before, during, and after the COVID-19 period (like in Juhász et al., 2023). This is important, since social and mobility networks adopt to external shocks (Napoli et al., 2023, Ubaldi et al., 2024) that could help us understand how policy interventions can be used to improve them. A dedicated Work Package on Data Management will be devoted to maximizing the overlap of analysed periods and to harmonizing the number of examined days and months that will depend on city-level specificities of mobility-restrictions during the pandemic, our negotiations with data providers, and the input of major stakeholders and Co-operation partners.

Special attention will be given to not harm any of the vulnerable populations and to manage sensitive personal data in a GDPR compliant framework. Segregated communities can become more isolated and demographic groups can be left out by mechanistic applications of the "15-minute city" concept. Our research will attempt to find solutions to minimize their risk of spatial isolation during proximity-oriented developments.

4.2. Overall project type

We are aiming cutting-edge strategic research answering how neighbourhoods in cities can be more "complete" and more attractive for various socio-economic groups and how public transport can be incorporated in the "15-minutes city" developments in Europe. We are using fine-grained mobility data to directly observe amenity visits and use them to quantify the complementarity of amenities in neighbourhoods. This technique will enable us to characterize neighbourhood trajectories and their resilience to external shocks; thus, will contribute to effective proximity-oriented developments. Additionally, we are combining data collection about street visuals to assess urban perception of specific neighbourhoods and to understand how their developments can increase active mobility of gender groups and facilitate social mixing. Further, highly policy-relevant areas will be addressed and linked to the "15-minute city" initiative for the first time: both roles of public transport, and work-from-home has been disregarded in the related literature so far, despite their importance in car traffic reduction.

COLINE is a research-oriented proposal and will focus on basic research. We will concentrate on producing new knowledge following rigorous scientific standards to better understand urban mobility and the potentials of the "15-minute city" concept in a more comprehensive way than it was done previously. Our focus will be on collecting, organizing, and processing data across cities, on developing innovative methods, on developing and testing theories.

We are aiming interdisciplinary science, combining various computational and social science fields that deal with urban environments. Data and network science will provide us with the toolset that have been used increasingly in urban research to investigate social segregation, the efficiency of transfer systems in cities, and the economy. We are going to leverage generative AI techniques, a field that is revolutionizing analysis and increases efficiency of decision-making. Research questions, theories, as well as rigorous causality tests from economics and social sciences will be integrated into the project. In all, we are going to do computational social science that is often defined as mixture of information science and social sciences and covers the overlapping disciplines listed above. Finally, we are aiming to have an impact on the planning literature that is very influential in proximity-based urban development.

The main outcomes of COLINE will be open-access data sources, codes, presentations, videos, and scientific publications, and popular media outlets, in which we will share new knowledge with colleagues in academia, in the public and private sector, and with the wider public. As expected from high-quality science, we are aiming for reproducible research.

Besides the research-oriented focus, COLINE will aim to have direct impact on policy and company innovation in the field of urban mobility and data-driven decision-making. A close collaboration with public transport and urban planning authorities will ensure that aims, methods, and results of the research will be discussed and tested with stakeholders who can use our new insights. In this regard, using mobility and public transport data in the "15-minutes city" planning procedure, and the generative AI techniques might induce innovation in these public sector stakeholders. We will also create data and software infrastructures that can generate value for the stakeholder telecommunication companies. The mobility-based amenity-space can be a technique that these companies might want to commercialize in their location-intelligence services. These potential policy and commercial applications are still within the scope of our research-oriented approach.

In COLINE, we will make sure that our research-oriented approach builds on the knowledge and priorities of the urban society that we want to understand. We will build and manage participatory workshops, where dwellers can express their thoughts and preferences about specific neighbourhoods, how they should be linked to other parts of the city, and how they should look like. During these workshops, participants will be asked to reflect on modified street images that will help us characterize demands for development in a systemic way and to measure what attractive and safe neighbourhoods look like for various socio-demographic groups.

4.3. Added Value of International Co-operation

International collaboration will be carried out by globally recognized scholars in economic complexity, social data science, network science, urban mobility, social segregation, urban economics, and labour economics. Each objective will be addressed by separate work packages led by a researcher who is at the frontier of that area. Research groups at the co-applicant institutions will be responsible for the successful accomplishment of the research targets, reaching the milestones and producing the deliverables. However, these targets will be set by all members of the international consortium together and most research activities will be carried out in collaboration across the participant institutions. These efforts will ensure that the quantitative analyses will be enriched with local knowledge of participant researchers. Also, the cross-group collaboration on specific topics will improve the general applicability of the methods and approaches and will also boost the efficiency of the tools as a wider pool of expertise will be used.

Objectives I, *II*, and *V* will be targeted by a common effort of the international consortium, in which all participating institutes will contribute data from their city, and some will contribute methods, analysis, and implementation. Addressing *Objectives III*, and *IV* will also require international collaboration but can be accomplished by fewer members as well. The work packages will be built on each other, such that the methodological developments and explored urban phenomena that are produced in relation to *Objectives I* and *II*, will be exploited in *Objectives III* and *IV*. Finally, the generalization in relation to Objective V will benefit from the international collaboration as the cities of the consortium will be the first test of widely reproducible applications. The international consortium consists of many persistent bilateral collaborations across members but will be new as a team. For example, Prof. Di Clemente and Prof. Lengyel are co-authors, Prof. Karsai and Prof. Alessandretti have collaborated already, Prof. Hidalgo holds a position at the Toulouse School of Economics and at the Corvinus University of Budapest. Such composition of persistent and new collaborations is ideal, since team members already know how to work with most of the partners, but the consortium will also provide enough diversity such that new combination of knowledge will be fruitful.

Results from five major cities will provide the opportunity to generalize the findings and methods for further countries as well. Similarities and differences among these cities will challenge us to establish empirical and theoretical frameworks that can be applied in various contexts. All these monocentric cities have well-developed public transportation systems, which are crucial for reducing reliance on private vehicles and supporting the "15-minute city" concept. The physical size and degree of urban sprawl vary among these cities. Copenhagen and Vienna, for example, are relatively compact and have policies in place that limit urban sprawl, making them more conducive to the "15-minute city" concept compared to cities with extensive suburban development. Higher population densities in cities like Budapest and Turin can make it easier to implement the "15-minute city" concept because amenities need to cater to a concentrated population. However, it can also pose challenges related to congestion and pressure on public spaces. The commitment of city governments to sustainable urban planning principles varies. Copenhagen and Vienna, for instance, have been leaders in promoting sustainable urban mobility and land use planning that aligns with the "15-minute city" ideals. Cities like Toulouse and Turin are exploring innovative solutions and pilot projects aimed at promoting the "15-minute city" concept, such as developing new green spaces and pedestrianizing certain areas.

Budapest and Vienna are already strongly linked in joint projects and various channels of inter-city exchanges. COLINE will further strengthen the synergies between Co-operation partners and stakeholders and will create a cross-sectoral learning environment for further cities. By involving municipalities, public transportation, and cell-phone providers as well in both Vienna and Budapest, and municipality units from Copenhagen, Toulouse, and Turin, we aim to maximize inter-city learning between stakeholders and our team and to engage and explore potentials in the urban mobility modelling for proximity-based developments.

5. Key activities and work programme description

5.1 Programme description

To address the research Objectives and to reach the desired impact, the specific tasks of COLINE will be organized into nine Work Packages (WPs). These WPs will be led by a scholar, who has experience and a prominent track record in the topic, but members of the research groups will participate in the WPs that are either led by the PI of their research group or led by other Co-Applicants. WPs will follow a workflow that will generate a clear but synergistic labour division, but data and efforts will be shared across groups to maximize the potentials of scientific discovery and inter-city learning. To establish a workflow in the project, we distinguish *General*, *Method-focused*, and *Analysis-focused* WPs.

The *General* WPs include *Project Management* (WP1) that focuses on the start of the project, on the co-ordination of the acquisition of new resources and labour force, and on all general tasks that are needed to the successful completion of the project (e.g. management of deliverables). In this regard, *Data Management* (WP2) will focus even more on the support of collection, safe storage, and sharing of the data across research groups and developing the protocols to collaborate on sensitive data that cannot be shared outside protected data labs and servers. *Project Dissemination and Policy Impact* (WP8) will concentrate efforts to share the knowledge generated in COLINE with the academic and public audience and consulting with major public and private sector stakeholders and channelling in crucial insights from urban societies. *Knowledge Hub* (WP9) will deal with sharing our results and applications within the Driving Urban Transitions Partnership and with learning from other projects and experts active in the field. These WPs together will provide the necessary conditions for successful project implementation and impact.

The *Method-focused* WPs will include *Amenity space from mobility data and neighbourhood dynamics* (WP3) and *Urban perception* (WP4) that will use the data collected in COLINE to characterize neighbourhoods with amenity portfolios, mobility traces, and urban perceptions that will be mined with the use of generative AI tools. The outcomes of these WPs will be leveraged further in the *Analysis-focused* WPs. These latter will be used in *Multilayer networks of urban mobility* (WP5), in *Segregation in the "15-minute city"* (WP6), and in *Amenity visits and time spent around home and work* (WP7) that will focus on understanding the empirical questions specified among the Objectives of COLINE.

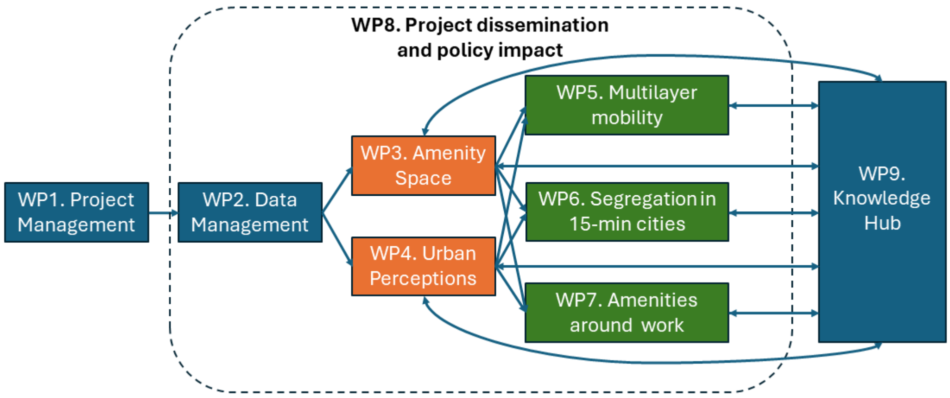


Figure 3. The tasks of COLINE will follow a workflow between *General* (blue), *Method-focused* (orange) and *Analysis-focused* (green) WPs. The project will aim to maximize the dissemination of results and policy impact that connects almost all WPs.

Each research Objective of COLINE will be addressed by at least one WP. *Objective I* will be addressed by WP3 and WP4. *Objective II* will be investigated in WP5; while *Objective III* will be looked at in WP6 and *Objective IV* will be targeted in WP7. Finally, *Objective V* will be aimed at with the help of many WPs that range from WP2 to WP9.

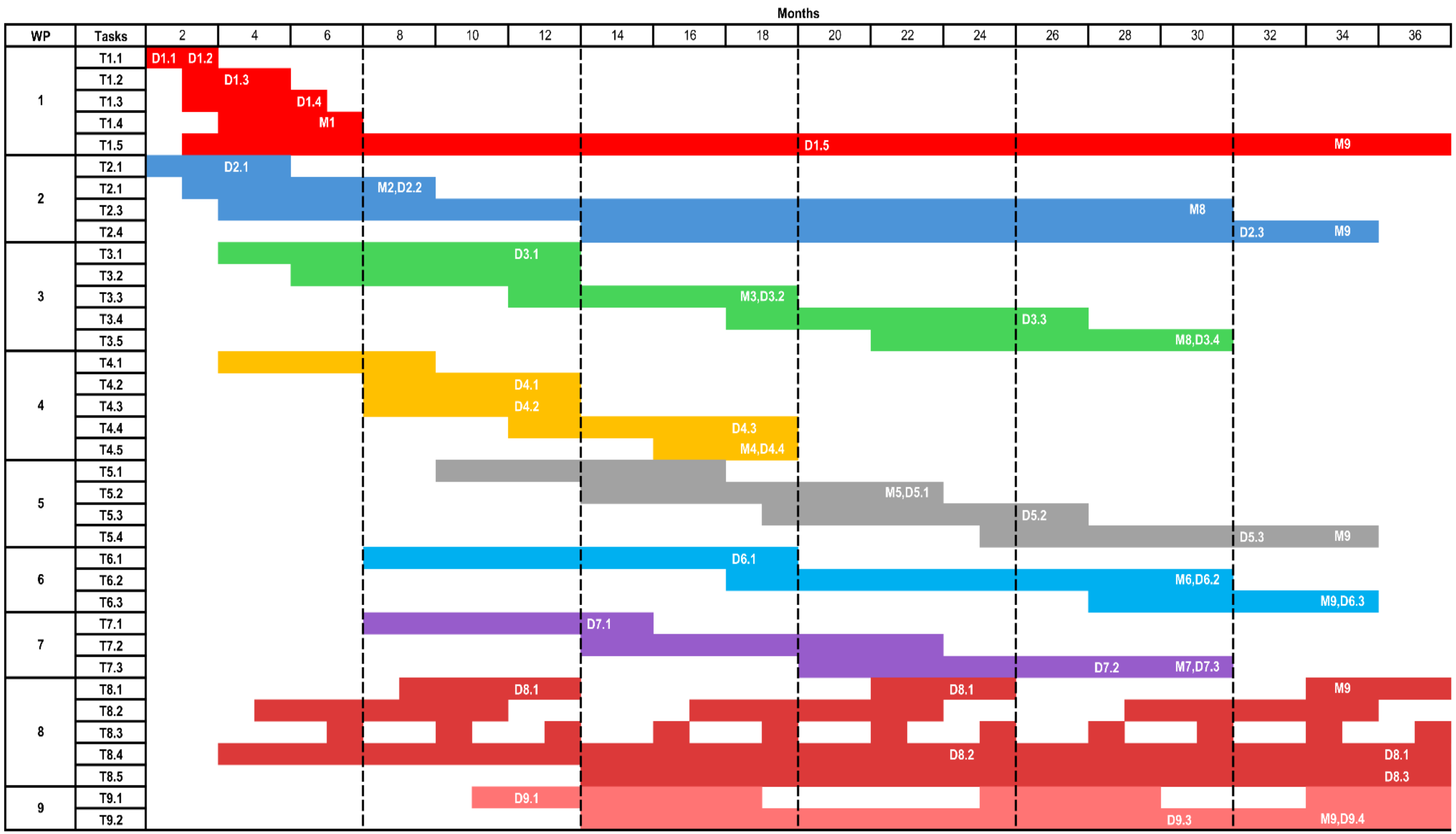


Figure 4. Gantt chart of COLINE. Rows represent tasks, while columns denote 2-month time intervals. Colours depict work packages. “T” stands for tasks, “D” denotes deliverables and “M” is short for milestones.

Table 5.1: Work package list

Work package No[2]	Work package title	Lead project partner No[3]	Lead project partner short name	Person-months[4]	Start month[5]	End month
WP1	Project Management	1	HUN-REN	5.6	1	36
WP2	Data Management and Ethics	1	HUN-REN	6.5	1	34
WP3	Amenity space from mobility data and neighbourhood dynamics	5	ISI	31	3	30
WP4	Urban perceptions	3	TSE	40	3	18
WP5	Multilayer networks of urban mobility	4	DTU	29	8	34
WP6	Segregation in the “15-minute city”	2	CEU	30	6	34
WP7	Amenity visits and time spent around home and work	6	CUB	23	6	30
WP8	Project dissemination and policy impact	3	TSE	11.5	2	36
WP9	Knowledge Hub	1	HUN-REN	5.8	9	36
			TOTAL		182.4	

Table 5.2: Deliverables List

Del. no.[6]	Deliverable name	WP no.	Delivery date[7]
D1.1	Kick-off meeting	WP1	Month 2
D1.2	Report on internal communication channels and online image	WP1	Month 2
D2.1	Data Management and Protection Plan	WP2	Month 4
D1.3	Deliverable management plan	WP1	Month 4

[2] Work package number: WP 1 - WP n.

[3] Number of the project partner leading the work in this work package.

[4] The total number of person-months allocated to each work package.

[5] Measured in months from the project start date (month 1).

[6] Deliverable numbers in order of delivery dates. Please use the numbering convention <WP number>.<number of deliverable within that WP>. For example, deliverable 4.2 would be the second deliverable from work package 4.

[7] Measured in months from the project start date (month 1).

D1.4	Report on recruitment	WP1	Month 6
D2.2	Data Collection Report	WP2	Month 8
D9.1	Knowledge Hub presentation 1	WP9	Month 12
D3.1	Access to cleaned GPS data	WP3	Month 12
D4.1	Curated data sample of streetscape images	WP4	Month 12
D4.2	Sample of AI generated images tailored to the participating cities.	WP4	Month 12
D8.1	Workshops and events 1.	WP8	Month 12
D7.1	Report amenity visits around home and work	WP7	Month 14
D3.2	Amenity space metrics	WP3	Month 18
D4.3	Online citizen participation events and promotion campaigns	WP4	Month 18
D4.4	Data handoff. about urban perceptions	WP4	Month 18
D6.1	Report on segregation and inequalities	WP6	Month 18
D1.5	Mid-term meeting	WP1	Month 20
D5.1	Method and corresponding open-source repositories to identify transport mode from GPS trajectories	WP5	Month 22
D9.2	Knowledge Hub presentation 2	WP9	Month 24
D8.1	Workshops and events 2	WP8	Month 24
D8.2	Campaign materials	WP8	Month 24
D3.3	Amenity resilience report	WP3	Month 26
D5.2	Report on public transport case study for the city of Copenhagen	WP5	Month 26
D7.2	Report on the analysis of neighborhoods by amenity mix, occupation mix and perceived safety	WP7	Month 28
D9.3	Knowledge Hub synthesis of results	WP9	Month 30

D3.4	Case study report on amenity visits	WP3	Month 30
D6.2	Report on segregation improvements through the "15-minute city" concept.	WP6	Month 30
D7.3	Database on work-from-home potential, amenity access at home, and amenity access at work by occupation	WP7	Month 30
D5.3	Synthesis of results on multimodal mobility.	WP5	Month 32
D2.3	Data Sharing Report	WP2	Month 32
D6.3	Policy recommendations to mitigate segregation	WP6	Month 34
D9.4	Communication to academic community and stakeholders	WP9	Month 34
D8.1	Workshops and events 3	WP8	Month 36
D8.3	Academic publications	WP8	Month 36

Table 5.3: List of milestones

Milestone number	Milestone name	Work package(s) involved	Expected date [8]
M1	Project image, Consortium Agreement, channels, and teams	WP1	Month 6
M2	Data access	WP2, WP3	Month 8
M3	Amenity space from mobility	WP3	Month 18
M4	Urban perception data and participatory workshops	WP4	Month 18
M5	Multimodal mobility framework	WP5	Month 22
M6	Improvement of segregation in multimodal mobility and by urban perceptions	WP6	Month 30
M7	Amenities around work	WP7	Month 30
M8	Dissemination and impact	WP1, WP2, WP5, WP6, WP8, WP9	Month 34

[8] Measured in months from the project start date (month 1)

Table 5.4: Work package description

Work package number	WP1	**Start date or starting event:**			Month 1	
Work package title	Project Management					
Project partner number	1	2	3	4	5	6
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB
Person-months per applicant:	3	0.3	1	0.5	0.5	0.3

Objectives

This WP aims to ensure effective project execution, including risk management, quality management and sound management of financial resources. The WP will support community building within COLINE, enabling a positive and safe work environment for all participants. It will ensure requirements of the Partnership and the relevant Funding Agencies are met timely and in good quality.

Description of work

Responsibilities in this work package: overall project management, including risk management, quality management, financial control, internal communication, and project administration. The management of COLINE is based on the joint efforts of the Main Applicant who will be responsible for the international coordination of the project and the Management Board (MB) that will include other Co-Applicants. WP1-leader will ensure that MB meets regularly and is properly informed of any developments within the project; minutes of the MB meetings will be included in our project administration. Each beneficiary and each Work Package leader will have to report on a regular basis to the MB. The Main Applicant will manage communication with the Consortium Board and the Advisory Board.

T1.1 - Internal communication channels and meetings. We will establish digital communication channels (eg. on Slack) that will enable us to talk about tasks collectively and in groups. We organize internal meetings for the whole consortium.

T1.2 - Deliverable management plan. A roadmap will be outlined for creating major deliverables that need collaboration across partners, with quality checks integrated to ensure the highest standards.

T1.3 - Online image. A project logo and website will be created.

T1.4 - Workforce recruitment coordination. New colleagues will be hired by the Main Applicant and by Co-applicants. The process will be coordinated to maximize the reach of the calls.

T1.5 - Risk and financial management. The MB will regularly discuss progress in WPs and will decide necessary corrections in the Deliverable management plan and monitor the financial management of Co-applicants.

Participants: HUN-REN KRTK will co-ordinate the Project Management WP that all Co-applicants will join.

Deliverables

D1.1 - Kick-off meeting. An online meeting including all Co-Applicants and Cooperation Partners. (Month 2, M1)

D1.2 - Report on internal communication channels and online image. The report will summarize the development of the online image and communication channels. (Month 2, M1)

D1.3 - Deliverable management plan. The plan will specify timing and tasks of participating institutions in producing the deliverables of all WPs, including risk management, and quality control. (Month 4, M1)

D1.4 – Consortium Agreement and report on recruitment. The agreement will be signed by all partners. The report will summarize the hiring process of new colleagues. (Month 6, M1)

D1.5 - Mid-term meeting. An online meeting including all Co-Applicants and Cooperation Partners. (Month 20, M8)

Work package number	WP2	**Start date or starting event:**				Month 1	
Work package title	Data Management and Ethics						
Project partner number	1	2	3	4	5	6	
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB	
Person-months per applicant:	3	0.5	1	0.5	1	0.5	

Objectives

COLINE will collect and process large data sources that might be sensitive and protected. Therefore, securing safe data access for participating institutions is of high priority. At the same time, our goal is to facilitate inter-city learning within the open-data framework. Therefore, we will generate data sets where observations will be aggregated and anonymized and thus, can be shared with the public.

Description of work

In this WP, we will carry out all the work that is needed to collect, store, process and share data.

T2.1 - Data management and protection guidelines. We will write a plan that will summarize data protection principles and rules. In this plan, we will appoint a Data Protection Officer and the Ethical Committee.

T2.2 - Support in the collection of data. In this WP, we will provide support to coordinate the collection of openly available data sources (eg. amenity data, street networks, street images, GTFS), commercially available (eg. mobility data) and secured administrative datasets. Special focus will be devoted to coordinating with the Magyar Telekom (Co-operation Partner) in Budapest, Magenta (Observational Partner) in Vienna, and the Co-Applicants. Data collection and cleaning will be done in the Method-focused and Analysis-focused WPs to maximize applicability.

T2.3 - Data storage and access. Secure servers will be devoted to storing the data that can be shared within COLINE partners.

T2.4 - Data sharing. We will create aggregated and anonymized data sets that can be published.

Participants: The WP will be led by HUN-REN KRTK that will facilitate the communication in the consortium and support the collection, storage and sharing of data. All Co-applicants will participate in this WP.

Deliverables

D2.1 - Data Management and Protection Plan. We will describe the principles of data protection, processing and sharing and name the Data Protection Officer and members of the Ethical Committee. (Month 4, M2)

D2.2 - Data collection report. This document will summarize the process of all data collection, storage, protection, and access. (Month 8, M2)

D2.3 - Data sharing report. This report will introduce the type of data that COLINE generates for external use and explain how the data was published. (Month 32, M8)

Work package number	WP3	**Start date or starting event:**				Month 3	
Work package title	Amenity space from mobility data and neighborhood dynamics						
Project partner number	1	2	3	4	5	6	
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB	
Person-months per applicant:	5	0.5	2	0,5	22	1	

Objectives

COLINE aims to utilize Location-Based Services data (LBS) from mobile phone apps to discern patterns in amenity visits and focus on how these amenities are interconnected, forming a network that influences the urban fabric. COLINE will study how these visitation patterns correspond with neighbourhood topological features, including amenity type, rate, and accessibility within the city's layout. The study will also examine disruptions in urban mobility and amenity visitation, investigating the factors that drive changes in urban mobility and resilience. This research will provide insights into how variations in visitation behaviours are shaped by urban density and the diversity of amenity spaces.

Description of work

T3.1 - Data purchase and cleaning. We will decide the best dataset to acquire from phone companies or LBS companies to have the optimal spatial and time resolution. We will define the pipeline to clean the dataset procured.

T3.2 - LBS data validation and integration. Measurements derived from LBS will be compared against high-quality statistics from the census and travel surveys to assess the validity of LBS as a proxy for official census data. Particular attention will be devoted to identifying any under-representation of disadvantaged population segments in behavioural groups. LBS data will be integrated with public transport data, socio-economic and amenities data. Integrated data will be shared with teams working on WPs 5-6-7.

T3.3 - Amenity space and visitation patterns. We will utilize Sequential Pattern Mining and other Machine Learning techniques to analyse citizens' high-resolution travel diaries extracted from LBS. We aim to discover the relationships between diverse amenities and their functional dynamics throughout different times of the day. We will establish a novel framework that automatically highlights the temporal and spatial relation between the different amenities in terms of visitation patterns, defining an amenity visitation space that will be used In *T3.4, T5.2, T6.2, T7.2*.

T3.4 - Amenity resilience. We will identify events that caused disruptions in city mobility and amenity visitation patterns. We will explore the factors driving long or short-term urban mobility changes and resilience. For instance, during COVID-19, we will capture the topological features of the amenity space that foster resilience, decoupling how the citizens' changes in visitation behaviour relate to the urban density and amenity space diversity. This metric will be used in *T3.5* to illustrate areas of policy intervention to increase neighbourhood resilience.

T3.5 - Case study of neighbourhoods. Under the guidance of stakeholders, we will examine case studies of specific neighbourhoods in the studied cities. The objective is to determine whether the framework implemented in previous tasks can provide accurate and valuable metrics for policymakers to implement policies aimed at improving social mixing and amenity accessibility. This task will be carried out in collaboration *T5.4, T6.3*.

Participants: WP3 will be led by ISI who will co-ordinate contributions of HUN-REN, TSE, DTU, CUB, and CEU. The task of data purchase will include all participating cities where data is available.

Deliverables

D3.1 - Access to cleaned GPS data. The data will be used in WPs 3-5-6-7. (Month 12, M2)

D3.2 - Amenity space metrics. New metrics on mobility-based amenity space to characterize neighbourhoods. The metrics will be used in WPs 3-5-6-7. (Month 18, M3)

D3.3 - Amenity resilience report. Empirical results to illustrate the relationship between amenity space and neighbourhood resilience. (Month 26, M8)

D3.4 - Case study report on amenity visits. Policy-related results about the potentials to improve amenity visits. (Month 30, M8)

Work package number	WP4	**Start date or starting event:**			Month 3	
Work package title	Urban perception					
Project partner number	1	2	3	4	5	6	
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB	
Person-months per applicant:	1	1	36	1	1	0	

Objectives

The objective of this WP is to collect, analyse, augment, and distribute data on people's perception of streetscapes. This data will enable us to quantify the relationship between the perception of neighbourhoods, amenities and their attractiveness revealed by individual mobility. The measurement will allow for policy-related recommendations about neighbourhood improvement in terms of safety and aesthetic appeal for various socio-economic and demographic groups.

Description of work

The package aims to collect, augment, and analyse data on people's perception of streetscapes.

T4.1 - Collect a representative sample of streetscape and other images for participating cities. The collection component will include the deployment of online data collection efforts designed to learn people's preferences across streetscapes from the collaborating cities (e.g. Budapest, Toulouse, Copenhagen, etc.).

T4.2 - Classify the images using a combination of crowdsourcing and machine learning methods. This effort builds on platforms developed at MIT by some of the members of this team to collected and analysed urban perception data with the help of machine learning models (Place Pulse (Salesses et al 2013), Streetscore (Naik et al. 2016), and Streetchange (Naik et al. 2017).

T4.3 - Generating A.I. images of proposed neighbourhood changes. We will closely collaborate with Toretei SRL to modify collected street images such that they can improve the perception of safety and aesthetic appeal of citizens.

T4.4 - Demographic and socio-economic differences of urban perception. We will use the AI generated images in citizen participation efforts to understand differences in perception across demographic characteristics (e.g. age, gender) and to understand urban segregation. This package will also explore heterogeneity in perception (e.g. across age and gender), urban segregation in terms of perceived inequality, and citizen participation modules. The task will be carried out in collaboration with policy stakeholders in the participating cities who will be able to invite NGOs active in the field of urban segregation and neighbourhood revitalization.

T4.5 - Data access. Resulting datasets will be provided to researchers working on *T3.4, T6.3*.

Participants: WP4 will be led by TSE. Every participating city will be represented by at least one Co-applicant to maximize inclusion of citizens.

Deliverables

D4.1 - Curated data sample of streetscape images. A collection of open-access and cleaned data from participating cities. (Month 12, M3)

D4.2 - Sample of AI generated images tailored to the participating cities. Generated images will reflect specific needs about neighbourhood developments to improve amenities, aesthetics, access, and safety. (Month 12, M3)

D4.3 - Online citizen participation events and promotion campaigns. A series of workshops with participation of citizens, NGOs, and policy authorities. (Month 18, M3)

D4.4 - Data handoff about urban perceptions. Data to be shared with WPs 3 and 6. (Month 18, M3)

Work package number	WP5	**Start date or starting event:**			Month 8		
Work package title	Multilayer networks of urban mobility						
Project partner number	1	2	3	4	5	6	
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB	
Person-months per applicant:	4	1	2	20	2	0	

Objectives

Quantify multimodal mobility patterns across various scales—from local interactions to city-wide movements—as individuals engage with diverse amenities. The analysis aims to assess how effectively public transportation can mitigate one of the principal challenges associated with the 15-minute city model: the risk of urban segregation. By quantifying these dynamics, we seek to identify targeted structural interventions in multimodal urban transportation networks that enhance efficiency, foster social integration, and decrease environmental emissions.

Description of work

T5.1 - Transport mode estimation. Develop a novel method for precise transport mode estimation from GPS trajectory data collected and cleaned in WP3, improving upon existing methods that analyse characteristic features such as velocity and acceleration. The method will integrate additional data reflecting transport infrastructure and dynamics, such as GTFS data for public transport and cycling infrastructure data. A machine learning-based framework will be employed to classify trajectories into distinct transport modes.

T5.2 - Multimodal travel in 15-minute cities. Investigate multimodal mobility (walk, public transport, car) through the framework of multilayer transport networks. We will model the roles played by different factors on mode choice as individuals access amenities, including time and cost, but also cognitive costs and individual preferences, and will pay special attention to the amenity space measures specified in *T3.2*. We will study differences across different groups with respect to background characteristics such as gender, age-group, and socio-economic status. Causal analysis through event studies will be applied to study the effects of infrastructural changes to the public transport network on travel patterns.

T5.3 - Inclusive public transportation in 15-Minute Cities. The concept of 15-minute cities seeks to enhance accessibility by augmenting local services, potentially leading to increased urban segregation, and reduced social mixing. This task will explore the inherent trade-off between local accessibility and urban diversity. Further, we will develop models for public transportation systems that not only improve accessibility but also promote social interactions and integration across different urban communities. This task is closely related to *T6.3*, which will continue the effort to include more dimensions of potential policy interventions. Results of *T5.3* will be used in *T3.5* to investigate resilience of neighbourhoods.

T5.4 - Improve sustainability of multilayer transport network. We will identify and address gaps in the public transportation network to significantly reduce car traffic towards amenities, thereby enhancing the network efficiency and sustainability and reduce transport emissions.

Participants: WP5 will be led by DTU but every participating city will be represented by at least one partner.

Deliverables

D5.1 - Method and corresponding open-source repositories to identify transport mode from GPS trajectories. We will deliver a method to work on multimodal mobility using GTFS and LBS data. (Month 22, M5)

D5.2 - Report on public transport case study for the city of Copenhagen. Results of our method will be tailored to specific city needs. (Month 26, M8)

D5.3 - Synthesis of results on multimodal mobility. Results across participating cities will be summarized. (Month 32, M8)

Work package number	WP6	**Start date or starting event:**			Month 6		
Work package title	Segregation in the “15-minute city”						
Project partner number	1	2	3	4	5	6	
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB	
Person-months per applicant:	3	18	1	2	3	3	

Objectives

The first goal of this WP is to identify segregation patterns in mobility mixing and urban perception of socioeconomic groups; and to identify inequalities in access to different kinds of amenities by subgroups of the population. The WP aims to explore the effects of the “15-minute city” concept, in terms of amenity allocation, improvement in transportation access, or improvement of urban space. Our question is whether and how segregation induced by proximity-driven development can be mitigated. We will pay special focus on vulnerable minorities and to explore gender differences in mobility segregation and amenity access.

Description of work

Task 6.1 – Segregation in mobility mixing and urban perception. The aim is to establish the actual mobility mixing and urban perception segregation patterns characterizing the cities in focus. The first step is the socioeconomic enrichment of mobility data via the combination of inferred home locations and socioeconomic indicators, given by the data providers or census. Subsequently, the relation between the mixing of socioeconomic groups, the accessible transportation (WP5), the visited amenities (WP3), and the perceived urban environment (WP4) will be explored.

Task 6.2 – Inequalities in access to amenities and urban functions. Due to biased amenity distributions, urban segregation, and limited access to transportation, inequalities may emerge in the access and composition of amenities, urban functions such as public spaces for sport and leisure, and the applicability of the whole “15-minutes city” concept for different socioeconomic groups. In this task our goal is to quantify these inequalities across the cities in focus. We will employ clustering techniques to understand how different communities interact due to the synchronicity of daily activities (routines), geographical patterns, and socio-economic stratification. This task will integrate visitation patterns captured in *T6.1* with multi-layer spatial-temporal analysis to assess the temporal signatures of neighbourhoods, capturing feature similarities in relation to city mobility dynamics.

Task 6.3 – Improved segregation through the “15-minute city” concept. Following work in *T5.3*, this task will further explore the impact of designed policy interventions - amenity distributions, improved urban environment and transportation access - on socioeconomic segregation and access inequalities. Further, we will study how the “15-minute city” concept could disadvantage vulnerable groups through gentrification, altered mobility, or the emergence of unaffordable services. This task will study how city policies can be targeted to increase perceived and experienced safety for women and elderly populations in the city, through a better composition and access to essential amenities.

Participants: WP6 builds on data obtained in WP3 - Amenity space from mobility data and neighbourhood dynamics; and is strongly related to WP4 – Urban perception and WP5 Multilayer networks of urban mobility. CEU will lead this work package in collaboration with Project Partners 3, 5 and 10 and other Co-applicants and Co-operation Partners.

Deliverables

D6.1 - Report on segregation and inequalities. The report will summarize research results from Vienna and other cities. (Month 18, M6)

D6.2 - Report on segregation improvements through the “15-minute city” concept. The report will contain results on the impact of designed policy interventions on segregation. (Month 30, M6)

D6.3 - Policy recommendations to mitigate segregation. We will formulate suggestions to avoid disadvantaging vulnerable minorities. (Month 34, M8)

Work package number	WP7	**Start date or starting event:**			Month 6		
Work package title	Amenity visits around home and work						
Project partner number	1	2	3	4	5	6	
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB	
Person-months per applicant:	0	0.5	2	0.5	2	18	

Objectives

This WP aims to identify how amenity visits around the workplace are different from amenity visits around home. The objective is to investigate worker and neighbourhood characteristics for which commuting to work can extend access to amenities, thus can modify the “15-minute city” concept. We also aim to quantify how working from home can contribute to aims of decreasing urban traffic. This will be targeted by identifying the neighbourhood qualities in terms of amenity portfolios, perceived safety or aesthetic appeal that can increase time spent in the neighbourhood with active mobility for those who work from home and by investigating how working from home changes access to amenities for those living in peripheral neighbourhoods.

Description of work

T7.1 - Characterizing amenity visits around the workplace. Building on data from *T3.3*, in this task, we will compare the amenities that people often visit around their workplace with the ones that they often visit around their homes. The comparison will be made across individuals living and working in different urban geographies of the participating cities. The aim is to show how access to amenities widens with the commuting to work and for whom.

T7.2 - Work-from-home potential and dynamics of amenity mixes around the workplace. Using administrative data that contains all companies and their workers, we will quantify the proportion of those occupations that can be done remotely. Then, for workplaces, we will analyse the dynamics of amenity visits around the workplace in different scenarios that test the resilience of amenity portfolios, like mobility restrictions during the COVID-19 pandemic, or the closure of offices. By extending the analysis with urban perception data, we will investigate the relationship between perceived safety at the workplace, amenity portfolios and occupations, and will check how the relationship varies across gender.

T7.3. - Work-from-home and amenities around home. Using census data, we will estimate the work-from-home potential of residential neighbourhoods and compare this metric to changes in temporal population density during COVID-19 infection waves to validate that LBS and cell-phone location data can capture work-from-home. Then, we will investigate scenarios of how changes in the local neighbourhood, in terms of amenity portfolio or perceived aesthetic appeal, can increase time spent in the neighbourhood and time spent with active mobility. We will analyse the trade-off between decreased pollution and more limited access to amenities due to work from home and decreasing commuting.

Participants: WP7 will be led by CUB and will cover analyses in Budapest where protected administrative and census data are available. The participation of TSE and ISI will ensure the integration of mobility and urban perception data.

Deliverables

D7.1 - Report amenity visits around home and work. We will compare periods before, during, and after the COVID-19 pandemic. (Month 14, M7)

D7.2 - Report on the analysis of neighbourhoods by amenity mix, occupation mix and perceived safety. The report will cover Budapest and other cities. (Month 28, M7)

D7.3 - Database on work-from-home potential, amenity access at home, and amenity access at work by occupation. Spatially aggregated data will be shared. (Month 30, M8)

Work package number	WP8	Start date or starting event:			Month 2	
Work package title	Project dissemination and policy impact					
Project partner number	1	2	3	4	5	6
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB
Person-months per applicant:	3	0.5	5	1	1	1

Objectives

The objective of the policy impact promotion and dissemination plan is to ensure our research findings influence urban planning policies and dialogue while fostering community engagement across the partner cities.

Description of work

T8.1 - Stakeholder Mapping and Engagement. During the stakeholder mapping we are going to identify policymakers, urban authorities, NGOs and businesses potentially interested in our project results to understand their needs and engagement levels. We will organize 3 workshops with local government officials to gather feedback and discuss preliminary findings.
T8.2 - Public Awareness Campaigns. Project developments will be regularly reported on the COLINE website and social media platforms such as Facebook, Instagram, LinkedIn, or X (formerly Twitter), highlighting each step and milestone achieved. Work with local partners to develop and execute multimedia social media and physical campaigns, including infographics, short videos, interactive webinars, data platforms and printed posters. We will collaborate with the involved cities to inform the public about the activities of COLINE and the benefits of compact, amenity-rich neighborhoods during their public events. In addition, we will organise informative webinars addressing the public.
T8.3 - Academic and Policy Conferences. Present research outcomes and findings at academic conferences, policy conferences, and public events.
T8.4 - Integration with Local Urban Planning Processes. Develop a long-lasting collaboration with local city planning and open data departments to integrate findings in the form of policy recommendations into upcoming urban development plans and public policy formulations.

T8.5 – Discussion with European Council of Spatial Planners. CUB will participate in Council meetings and will coordinate an event to disseminate results among a wide European network of planners and learn potential areas of policy-relevant research from Council members.

T8.6 - Scientific articles. Publish research results in peer-reviewed, global leading interdisciplinary and field journals.

Participants: The work-package leader TSE will coordinate the contributions of all Co-applicants who will engage with the tasks.

Deliverables

D8.1 - Workshops and events. Workshops with local stakeholders, academics, and community members. (Months 12, 24, 36, M8)
D8.2 - Campaign Materials: A suite of digital and print materials designed for public participation and awareness. (Month 24, M8)
D8.3 - Academic Publications. Policy relevant academic papers. (Month 36, M8)

Work package number	WP9	**Start date or starting event:**			Month 9		
Work package title	Knowledge Hub						
Project partner number	1	2	3	4	5	6	
Project partner short name	HUN- REN	CEU	TSE	DTU	ISI	CUB	
Person-months per applicant:	2	0.5	2	0.6	0.5	0.2	

Objectives

Participation of a nominated expert in the DUT Knowledge Hub community building. Sharing knowledge generated in COLINE and learning from other DUT projects.

Description of work

T9.1 - Participation of COLINE in the DUT Knowledge Hub. We will devote time and budget to participate in the Knowledge Hub events to share results and learn from other projects.

T9.2 - Collaboration with the Knowledge Hub Management and the 15mC Transition Pathway Programme Management. We will work to synthesize results across DUT projects and communicate results to the scientific community and to stakeholders.

Participants: HUN-REN and TSE will devote most time to WP9 but each Co-applicant will participate.

Deliverables

D9.1 - Knowledge Hub presentation 1. We will summarize project aims and first steps of Milestones 1 and 2. (Month 12, M9)

D9.2 - Knowledge Hub presentation 2. We will present the results of Milestones 3-4-5. (Month 24, M9)

D9.2 - Synthesis of results, collaboration with other partners. We will work with the DUT team. (Month 30, M9)

D9.3 - Communication to the scientific community and stakeholders. We will present results and write short summaries. (Month 34, M9)

Table 5.5: Risk mitigation

WP	Risk	Mitigation
WP1	One or few Co-operation partners will lose interest due to unforeseen external events.	COLINE will be carried out in five cities that will mitigate the risk of unsuccessful completion due to eventual exits of partners.
WP2	Data providers may fail to deliver the purchased datasets. Data collection is failed or delayed.	We aim to use open access data for the planned research. We are in contact with multiple data providers who may provide alternative data solutions.
WP3	The obtained GPS data may be too sparse for a meaningful analysis. Similarly, data from cell towers of mobile phone companies may be not precise enough for identifying amenity visits.	We work with test data to understand the richness and precision of GPS and cell-tower data. If necessary, we will apply a higher aggregation level in terms of time and geography. If necessary, we contact additional data providers. If necessary, we construct the amenity space aggregating subsequent amenity visits taking together all participating cities into one pool.
WP4	Participation of citizens in the urban perception workshops will be small or biased.	We will work with local urban authorities to increase participation rate and identify communities and NGOs who are interested in the objectives of the WP.
WP5	The different layers of multimodal mobility might be difficult to match between different data sources. Further risk is due to heterogeneous data sparsity.	If necessary, data will be aggregated in terms of time and geography to find the optimal resolution that enables the match between mobility layers.
WP6	The main risk of this WP is its dependency on data collection in terms of mobility, amenities, and perception. Further risk is the unavailability of corresponding socioeconomic maps.	Dependency on data collection will be mitigated by identifying alternative data sources. The lack of socioeconomic information will be solved by using higher level aggregated maps. Alternative solution would be to focus the analysis on cities where all the data is available.
WP7	Aggregate indicators of home locations from census data might not be exported from the secure data lab due to the decision of the Statistical Office that is still unknown.	We will import aggregate mobility networks into the secure data lab and continue the WP within the data lab.
WP8	Ineffective dissemination of project findings could limit impact and awareness	A comprehensive communication plan will be developed. Regular updates will be provided to stakeholders. Diverse dissemination channels, including social media, webinars, and public forums, will be utilised to ensure broader reach and engagement.
WP9	Failure of building connections with earlier DUT project PIs.	DUT Principal Office is hosted in one of the partner cities, that will help enormously to reach out to former DUT PIs.

6. Impact of the project

6.1. Expected impacts

COLINE will have five major outcomes that will be likely of direct application value for public sector and urban government authorities regarding the “15-minute city” concept and directly or indirectly impact major stakeholders in participating cities (Table 6.1). All these outcomes will aim to decrease car traffic and increase active mobility and public transport usage and foster healthy and inclusive neighbourhoods that will benefit the entire urban society.

Table 6.1. Impact on stakeholders in participating cities. ** indicates direct, while * denotes indirect impact. The estimates of impact are based on discussions with Co-operation partners.

	Urban authorities	NGOs	Private Companies	Citizens
1. Measurement of complete neighborhoods	**		**	*
2. Urban perceptions to increase active mobility	**	*	*	*
3. Public transport in the “15-min city”	**			*
4. Inclusion of vulnerable groups in the “15-min city”	**	**		*
5. Commuting to work	**		*	

First, COLINE will provide a new measurement of the completeness of the amenity portfolios in neighbourhoods, based on actual mobility of urban dwellers. This new approach will extend the methodological toolset of recent policy work of amenity accessibility that is based on walking- or biking distances. Leveraging large mobility datasets, we will trace amenity visits that will allow for the quantification of amenity demand and its urban geographies on a large scale. The method will enable us to generate data-driven policy recommendations about place-based developments. We will be able to tell what missing amenity types should be located to what neighbourhoods in the city to serve local demand. We aim to generate amenity-spaces for cities, based on co-visitation of amenities, an abstract construct that could generalize the methodology to other cities that we will not be able to investigate in detail. The method will also allow us to measure the resilience of amenity portfolios against external and economic shocks.

These advances in methodology will be likely of value for city authorities that prepare “15-minute city” development plans including support for amenity-rich neighbourhoods that are resilient against external shocks. Citizens will also indirectly benefit from such neighbourhoods that could serve more of their demand within proximity of their home. Further, the new method will generate business value for the participating cell-phone companies that are providing location intelligence services to estimate local demand for partners. We will provide support to increase the value of their product.

Second, measuring whether locations are perceived safe and attractive will enable us to provide recommendations for urban developers to improve their city. Leveraging generative AI techniques, we will run participatory workshops in which citizens will be able to express what developments would increase the safety and attractiveness of specific locations. This effort will help stakeholders understand how active mobility can be induced in neighbourhoods by developing safe accessibility in their streets. Similarly, showcasing how developing the aesthetics of locations improves the attractiveness of neighbourhoods. This effort can not only impact active mobility but also can help to make neighbourhoods inclusive where various socio-economic groups can mix.

Besides the direct policy impact, our contribution will be beneficial for NGOs that we will map in collaboration with city authorities and who are active in the investigated neighbourhoods or in helping vulnerable groups. Using our new tools, they can better understand how unsecure or unattractive areas could be transformed into inclusive and welcoming neighbourhoods. Naturally, local services and citizens will be able to enjoy the benefits of neighbourhood upgrading as well.

Third, COLINE will offer empirical tools and data to include public transportation into the “15-minute city” concept. We will show how public transport links across neighbourhoods can link demand for certain amenities and how changing modes of transportation within individual trips could contribute to reduction of car traffic in cities. These new methodological developments will be of value for public transport companies who can use the new knowledge to their collaboration with city authorities to increase the efficiency of public transport and to decrease congestion.

Improving public transport connections to amenities that are not available in the immediate neighbourhood will benefit citizens who otherwise must use their car. However, improving access by public transport will not only impact urban dwellers who live in the periphery of cities where only few amenities are available but will also upgrade living conditions in amenity-rich neighbourhoods where congestion is high.

Fourth, empirical measurements of segregation in the “15-minute city” will provide us with scenarios about the potential segregation outcomes that proximity-based urban development can induce. We will focus on vulnerable groups who might face the potential indirect challenges of polycentric development. Ethnic minorities who live in segregation, and demographic groups for whom active mobility is more difficult (for example women or elderly).

The research will directly benefit urban authorities and NGOs as the mitigation of urban inequalities and segregation is a common goal that is targeted by public bodies and the non-governmental sector. Indirect effect on the vulnerable population will be aimed by having impactful conversations with policymakers who can initiate important changes.

Fifth, we will investigate the amenity visits around the workplace and compare that to amenity visits around home. This will feed our conversation with urban public authorities to better understand how much reduction of car traffic can be achieved by proximity-based developments around residential areas. Further, neighbourhoods that host many workplaces should be made resistant against external shocks like mobility restrictions or industry-level crises. Thus, our research can help local services who build their existence on the demand generated by employees who visit them during the day.

Besides the local impact in participating cities, we aim to achieve global implications too. The project will contribute to the ongoing scientific discussions on the possibilities and consequences of “15-minute city” developments around the globe. We will develop and combine techniques that are of great importance in research in the fields of urban mobility, planning, sociology, urban economics, and data science.

We expect to have implications in other European cities, where public transportation plays a great role in reducing car traffic. Like in the case of other methodological developments that have been published before by members of the consortium (Hidalgo et al., 2007, Hidalgo and Hausmann, 2009), we expect that researchers and practitioners will use the new approaches of COLINE.

6.2. Relevance and contribution of the project to the goals of the call

COLINE aims to present methodologies and establish new partnerships between the researchers, public authorities, and private companies to address key problems that are explicitly outlined in Transition Pathway 15mC Topic 2: “Mobility and Planning Policies for Proximity-oriented Developments”. We aim to better understand the potentials and limitations of the “15-minutes city” developments in participating cities and elsewhere in Europe and on the globe. In particular:

- As a major contribution, we will use mobility data to characterize the completeness of neighbourhoods that will be able to capture the revealed demand for services and the use of urban space in the everyday life of inhabitants.

- The amenity space based on mobility will enable us to investigate future scenarios of how neighbourhood centres can react to external shocks and deliver insights about desired amenity portfolios in neighbourhoods.

- We will measure the perception of citizens about streets and neighbourhood centres with AI-augmented participatory efforts that will help us understand how improvements of safe accessibility can increase the attractiveness of local centres.

- The insights learned from urban perception measurement will inform policy on how neighbourhood centres should be developed and regulated to increase their attractiveness for various socio-economic groups.

- We will develop a multimodal mobility approach in which public transportation will be understood as a measure to decrease car use and emissions. Mobility with public transport will be combined with active mobility to see how increasing access to amenity rich neighbourhoods can favour the resilience of dense and peripheral parts of cities.

- To better understand how the six urban functions contribute to the feasibility of the “15-minute city” concept, we will compare amenity visits around home and work. This will help us better understand what the “15-minute city” can offer for working and learning, being core urban functions, besides the relatively better researched functions of living, enjoying, caring, and supplying.

- We will reflect on potentially emerging issues of “15-minute city” developments like increasing segregation and the lack of inclusion of vulnerable demographic groups. We will assess potential strategies that can minimize these unintended consequences of 15mC policy.

- Finally, with all the above innovative contributions at hand, we will investigate how the resilience and thrive of neighbourhoods depends on several aspects of the 15mC including segregation, safety perception, public transport, commuting to work, or working from home.

To advance the topics of the call, COLINE will harness the expertise of an interdisciplinary team of international experts. We will collaborate with key stakeholders in participating cities, fostering new links among urban authorities, public transport entities, mobile phone companies, and transport planners. By evaluating urban perceptions and the impact of proximity-based development, we aim to inform policies by gauging citizen opinions on neighbourhood centres in both densely and sparsely populated urban areas. The varied city sizes involved ensure insights will be drawn not only from large cities but also from medium-sized cities, where the “15-minute city” concept may differ.

Overall, COLINE will conduct data-driven analyses on how “15-minute city” policies and the enhancement of polycentric cities correlate with active mobility, considering lifestyle diversity and various data sources that capture the complexity of urban socio-economic systems. This approach will deepen our understanding of how proximity-driven developments can be realised effectively while minimising potential adverse effects.

6.3. Dissemination and/or exploitation of project results, and IP management

COLINE aims to provide solutions that are applicable across a series of cities. This will be achieved through a multi-level engagement with urban authorities, data providers, planners, citizens, and the academic community.

The insights that we gain from the collaboration with municipalities, public transport, mobility planning, and telecommunication companies will help us maximize the practical exploitation of research efforts. We will work with urban authorities responsible for mobility planning in all participating cities and with the same set of stakeholders in Budapest and Vienna. In the latter case the Wiener Linien (public transport company), the urban planning department of the Municipality of Vienna, and Magenta (mobile-phone provider, part of Deutsche Telekom) will be observers of the project, which will mean that we can contact them for asking advice and their offer for data service. These inter-city synergies will create an environment that includes inter-city learning from the beginning of the project. For example, the practice of tracing individual visits to amenities with mobile-phone data in a GDPR-compatible way will be developed between the co-applicants and Magyar Telekom in Budapest and then will be shared between mobile-phone providers. Furthermore, we will work with the Hungarian Society of Urban Planners and represent the project in the European Council of Spatial Planners to establish a close connection to a diverse audience of urban policy practitioners. Four policy-oriented workshops over the project timeframe will be organised to share results with a wide audience of European city authorities. During these workshops, further steps will be discussed with Co-operation Partners, and we will collect comments and insights from other cities too.

We will also closely engage with citizens and with the General Public. By measuring the perceptions of citizens about the aesthetics of neighbourhoods, their accessibility, and safety in AI-augmented participatory workshops, we will learn what urban dwellers think about places and how they think these should be developed. We will develop a project website to acquaint the general audience with the project, its initiatives, and the collaborating partners. A dedicated section on the website will showcase periodic reports on COLINE's activities and project outcomes. It will feature concise extracts and visualizations of results from various work packages, catering to broad interest. Additionally, we will establish COLINE social media accounts to enhance project outreach. Press releases and public events in participating cities will increase engagement with the local community.

To enhance public engagement, COLINE will access open data from a wide range of European cities, including public transport networks, street views, and amenities. All datasets and the code used for replicating data sources and analyses will be published in open-access journals and repositories. COLINE plans to initiate a platform that will enable the visualisation of mobility maps for the cities studied. This platform will feature interactive and responsive elements, allowing users to choose specific data sources for map visualisation. It will facilitate easy comparisons of different quantities on a map and navigation across various locations. Consortium members will seek additional funding to launch this project, which will extend the fundamental research undertaken by COLINE.

COLINE will follow the open-access initiative and publish intellectual property under Creative Commons license. Intellectual Property of participating organizations will not be transferred to the consortium. The dissemination of the scientific outcomes from the project's work packages through high-impact scientific channels. These outcomes, ranging from novel algorithms to empirical findings, will be submitted to prestigious, broad-spectrum journals such as Science Advances, Nature Communications, and PNAS, to new interdisciplinary journals with a related scope like Nature Cities, and to leading field journals. To reach a wider audience, we will also craft scientific dissemination articles. Given the project's multidisciplinary nature, COLINE will seize opportunities to present research outputs at prominent international conferences, including conferences of the Complex Systems Society and Network Science Society, the American Association of Geographers, the European Regional Science Association, and at the International Conference on Computational Social Science.

6.4. Gender and diversity aspects

Inclusion is a strong component of the "15-minute city" concept. Accordingly, COLINE will investigate the aspects of inclusion in terms of socio-economic inequalities and segregation, and inequality in active mobility in terms of gender and age.

Gender differences have been reported in terms of active mobility (Gauvin et al., 2020, Collins et al., 2023). In the context of urban infrastructure and perceptions, women are less likely to choose active mobility if they don't feel safe to ride a bike (Battiston et al., 2023) or the neighbourhood feels unsafe (De Nadai et al., 2016). The inequality in mobility can lead to unequal social capital and opportunities (Ilyés et al. 2023, Lőrincz and Németh 2022). Therefore, we will investigate how proximity-based developments can increase active mobility of women by making streets and bike lanes safer and the neighbourhoods of amenities more secure. This will be achieved by participatory workshops where street images will be altered by generative AI and we will survey the perceptions of citizens to learn what changes could increase their sense of security and inclination to walk or bike in the area. These measures will be extended to understand the developments needed to make neighbourhoods feel safer for women during the day and at night. Finally, we will investigate the question how multimodal mobility, and the mixed use of active mobility and public transport can improve gender balance of amenity visits in central and also in peripheral areas of cities.

In the participatory urban perception measurement, we can also investigate other demographic groups who are less likely to choose active mobility or public transportation in dense urban environments. Families with small children or elderly citizens might drive even if their target is within walking distance or is easily accessible with public transportation. By inviting these groups to the urban perception workshops, we can ask how neighbourhoods should be developed in order to increase their engagement in active mobility. We will link these new pieces of information to the surveys that are regularly conducted by public transport companies to assess the satisfaction of their customers and cross-check whether improving public transport to amenities can substitute driving for these demographic groups.

Increasing segregation is one of the potential adverse effects of proximity-driven developments. Scholars are warning that the "15-minute city" can further separate socio-economic groups (Abbiasov et al. 2024). Residential segregation has been widely reported in many urban contexts, but decreasing mixing across neighbourhoods can hurt the abilities of urban areas to mix diverse knowledge that made cities the engines of economic and technological progress and be especially harmful for segregated communities - be them lower class or ethnic groups - whose economic opportunities are limited. Here we contribute to the ongoing discussion in the interdisciplinary urban science literature about socio-economic mixing. We will use mobility data, in which the home locations of users can be identified and, based on information on the block level, we can infer socio-economic status of individuals. Using this technique, consortium members have already identified those amenity locations where most socio-economic mixing takes place (Juhász et al., 2023). Here we will extend this work into several new directions that address the role of "15-minute city" in segregation. For example, we will investigate how amenity clusters in the overlap of walking distance from thriving versus lagging neighbourhoods can facilitate socio-economic mixing, can policy facilitate this process and by supporting which types of amenities? We will also analyse the role of commuting to work that is found to mix different socio-economic groups, but we do not understand how the amenity visits around work replaced by amenity visits around home impacts segregation. Finally, we will use the urban perception data collection to analyse how neighbourhoods and amenities could be made more attractive for diverse socio-economic groups.

The project consortium is diverse in terms of gender. Out of 14 partners, 8 are represented by male coordinators and 6 have female coordinators. Gender balance in the entire research team will be an important aim when hiring new postdoctoral researchers in the project.

7. Quality and efficiency of project implementation

7.1. Consortium resources

The project consortium comprises globally recognized leaders in mobility analytics, urban data science, urban economics, and computational social science, possessing all scientific and technical competencies required for this research. These competencies include measuring neighbourhood completeness by linking mobility traces to amenities (Di Clemente, Karsai) and constructing amenity space (Hidalgo, Lőrincz), analysing urban perceptions and their impact on mobility (Hidalgo), multimodal transportation networks (Alessandretti), social mixing in neighbourhoods (Karsai and Di Clemente), commuting patterns (Lengyel and Lőrincz), and advanced approaches of multimodal mobility (Alessandretti). The team has experience collaborating with policymakers to support urban development (Hidalgo) and address major societal challenges, such as mobility restrictions during pandemics (Karsai). Coordination with a diverse range of stakeholders is managed by the lead applicant in Budapest (Lengyel). To conduct empirical work, six postdoctoral researchers will be hired, contributing a total of 182.4 person-months, supported by the co-applicants. Consortium partners have access to necessary data and infrastructure. The major applicant's institution (HUN-REN Centre for Economic- and Regional Studies) is experienced in coordinating European projects and has a dedicated Project Management Office for administrative tasks. All co-applicants have administrative support for national funding and reporting. The consortium's diverse scientific approaches in fields like economic complexity and network science will foster an interdisciplinary learning environment.

City authorities, public transportation companies, large telecommunication corporations, and SMEs active in urban innovation and planning will be closely engaged in the project as Co-operation partners. Frequent meetings will be held to discuss aims, empirical details, and results of the WPs. Project summaries will be presented for an audience including management of Co-operation partners and the wider community of European city authorities. The Municipality of Budapest, the Copenhagen Commune, Toulouse Metropole, Urban Innovation Vienna, and 5T from Torino will be invited to comment on the empirical design of all empirical work packages that address amenity visits with mobility data and that aims to exploit that measurement of urban perceptions. They will be able to formulate specific questions to adjust the research to address problems that are important for the “15-minute city” policy in their respective cities, and particularly for the development of neighbourhoods. We will work in close cooperation with telecommunication companies in Budapest and Vienna – Magyar Telekom in Budapest and Magenta in Vienna (as an observational partner) – to define amenity visits using their mobility data. Together with them, we will discuss the legal and technological possibilities to construct such data that we can analyse and share. Public transport companies in Budapest and Vienna – BKK and Wiener Linien (as an observational partner) – will be invited to discuss the empirical work on multimodal mobility and the role of public transport in the “15-minute city”. We will ask them to share the rich data that they collect on mobility including public transport vehicles, streets, and bike lanes.

Toretei, an SME focusing on urban innovation and street design, will work with us on the urban perception measurement as a Co-operation Partner and a potential subcontractor. They will help us organize the participatory workshops and modify street images with their generative AI technology to learn which local changes in neighbourhoods’ aesthetics and safety perceptions can facilitate active mobility.

Every Co-operation Partner will have the opportunity to request adjustments in the empirical design to address issues in the specific urban contexts they aim to tackle. Observational Partners have already expressed interest in collaboration and will be contacted to establish a partnership for the usage of their data in the project. Contact persons at each partner will be able to communicate the research within their organization.

7.2. Management structure and procedures

COLINE will co-ordinate a complex research program carried out by six Co-Applicants in five countries. One of the major objectives of the project is to facilitate inter-city learning and to generate new knowledge that is applicable in a series of cities. Therefore, we will conduct a synergistic research agenda across the participating cities that requires intense co-ordination of the research activities of groups.

The management structure can be described by 6 categories of participants with distinct responsibilities:

1. The *Main Applicant* will be responsible for the general success of COLINE. HUN-REN will carry out the international co-ordination of the project and lead the related activities of project and data management and Knowledge Hub work packages (WP1, WP2, WP9).

2. *Co-applicants* will lead methodological and empirical work packages (WP3-7), in which the objectives of COLINE will be addressed. The Co-applicants will have double responsibilities: 1. they will lead their groups and fulfil administrative duties as requested by their national funding agencies, and 2. They will co-ordinate the international research related to the WP that they lead. Decisions that will be made by Co-applicants and their teams will typically cover hiring, small-scale subcontracting, and operational decisions about empirical design.

3. The *Management Board* will consist of the Main Applicant and the five Co-applicants and will be the platform to share local experience and to discuss strategically important questions related to the international scale of the project. The Management Board will decide on questions that require intercity synergies, such as data purchase and storage, large-scale subcontracting, and workshops.

4. The *Consortium Board* will include every contact person from all consortium members. This group will meet less frequently during the workshops planned in WP8 and will discuss the policy-relevant aspects of COLINE and will make strategic decisions about the project.

5. We will invite global leader scholars in the field of urban data science, economics, sociology, transportation, and planning to the *Advisory Board*. This group of experts will be invited to comment on the academic aspects of the project to ensure that our activities will contribute to the latest developments of these fields.

6. We will work with an *Ethical Committee* and appoint a *Data Protection Officer* who will oversee all decisions about the data-related ethical questions.

The above management structure will ensure that every day and more strategic decisions as well will be made at the most appropriate level to remain efficient but provident too. This is crucial given the scale and complexity of COLINE.

7.3. Ethical and regulatory considerations

COLINE will collect and process large data sources that might be sensitive and protected. Therefore, securing safe data access for participating institutions is of high priority that we will carry out in WP2 "Data Management and Ethics". T2.1 in WP2 aims to establish the "Data protection and management guidelines" that will set the rules for data protection, access, and publication, following GDPR and national law. We will describe these principles in the Data Management and Protection Plan (to be delivered in Month 4). This document will synchronize the measures of data protection that are already at place in the participating organisations.

In the Data Management and Protection Plan, we will appoint an Ethical Committee that will be in charge of giving permission to the collection, storage, access, and analysis of personal data. The document will also appoint a Data Protection Officer who will oversee monitoring the data-related processes of COLINE.

Data will be stored on secure servers, and access will be limited to those participant researchers only who sign the requested documentations. In some cases, for example in the case of census data, access to the data will be possible in safe and protected data rooms that are controlled by national authorities. All publications, including technical reports, scientific publications will not enable third parties to identify individuals in the data that we analyse. In case of data publications, we will aggregate observations into a time and geography resolution that does not allow for identification.

7.4. Individual project partners

Project Coordinator - Balázs Lengyel, HUN-REN Centre for Economic- and Regional Studies

The HUN-REN Centre for Economic- and Regional Studies (HUN-REN) is a leading research institute dedicated to the study of economic and regional development in Hungary. It focuses on high-quality empirical research to inform policymaking and contribute to the scientific understanding of complex economic and social dynamics. The main task of HUN-REN will be the leadership of COLINE and participation in data collection, and analysis. HUN-REN has experience in H2020 consortium leadership, in urban data analytics and analysis of economic and social networks.

Balázs Lengyel is an Associate Professor and Senior Research Fellow and leads the ANETI Lab at HUN-REN and at the Corvinus University of Budapest. He is an economic geographer and aims to understand how social interaction facilitates economic and technological progress embedded in geographical space. Balázs is an expert in urban networks. He has initiated the Budapest Mobility Workshop that brings together major public and private stakeholders to investigate social aspects of urban mobility. Portfolio: ORCID; Google Scholar.

Education
2020 Habilitation, Economics, University of Szeged
2010 PhD, Economics, Budapest University of Technology and Economics
2004 MSc, Economics, University of Szeged

Current Positions
2023- Associate Professor, Corvinus University of Budapest
2020- Senior Research Fellow at Corvinus Institute for Advanced Studies
2017- Senior Research Fellow at the HUN-REN Centre for Economic and Regional Studies, Head of Agglomeration and Social Networks Lendület Research Group

Previous Positions (selected)
2013-2017 Research Fellow, HUN-REN Centre for Economic and Regional Studies
2010-2018 Research Fellow, International Business School Budapest

Selected Grants and Projects
2021-2025: NKFIH, 121K EUR,“Mobility, social- and economic networks, and inequalities in cities”
2018-2020: NKFIH, 54K EUR, “Spatial Diffusion of Innovation”
2017-2021: MTA, 404K EUR, ”Agglomeration and Social Networks”

Supervision: 2 PhD students at University of Szeged, 10 postdocs at HUN-REN, 4 visiting PhD

Selected Publications
Abbasiharofteh M, Kogler DF, Lengyel B (2023) Atypical combination of technologies in regional co-inventor networks. Research Policy 52(10) 104886.
Juhász S, Pintér G, Kovács Á, Borza E, Mónus G, Lőrincz L, Lengyel B (2023) Amenity complexity and urban locations of socio-economic mixing. EPJ Data Science.
Kovács Á, Juhász S, Bokányi E, Lengyel B (2022) Income-related spatial concentration of individual social capital in cities. Environment and Planning B 50(4): 1072-1086.
Bokányi E, Juhász S, Karsai M, Lengyel B (2021) Universal patterns of long-distance commuting and social assortativity in cities. Scientific Reports 11: 20829.
Tóth G, Wachs J, Di Clemente R, Jakobi Á, Ságvári B, Kertész J, Lengyel B (2021) Inequality is rising where social network segregation interacts with urban topology. Nature Communications 12(1): 1143.
Eriksson R, Lengyel B (2019) Co-worker networks and agglomeration externalities. Economic Geography 95(1): 65-89.

Staff members
Johannes Wachs, Research Fellow; *Bence Kovács*, Data Engineer; *Júlia Tarnai-Kiss*, project Assistant; *Róbert Bank*, Project Manager; *Sándor Juhász*, Research Fellow

Project Partner 2 - Márton Karsai, Central European University

The Central European University (CEU) is a distinguished graduate-level institution known for its research and education in the social sciences, humanities, law, management, and public policy, accredited in both the United States and Austria. The main task of CEU is the leadership of WP6, to participate in the analyses of other WPs, and to act as a national contact point for Austria. CEU has led and participated in several European grants.

Márton Karsai is an Associate Professor at the Department of Network and Data Science at the Central European University in Vienna where he leads the Computational Human Dynamics team. He is a network scientist with research interest in human dynamics, computational social science, and data science, especially focusing on socioeconomic status inference, mobility and social network segregation and segregation dynamics. He is an expert in analysing large human behavioural datasets and in developing data-driven models of social phenomena. Portfolio: ORCID; Google Scholar.

Education

2024	Doctor of Science (defended), Statistical Physics, Hungarian Academy of Sciences
2019	Habilitation, Computer Science, Ecole Normale Supérieure de Lyon
2005 - 2009	PhD, Statistical Physics, University of Szeged, Hungary; Univ. Joseph Fourier, France
2000 - 2005	MSc, Computational Physics, University of Szeged, Hungary

Current Positions

2019 - present	Associate Professor, Head of Department elect, Director of Social Data Science MSc program, Central European University, Vienna, Austria
2021 - present	Research Fellow, Rényi Institute of Mathematics, Budapest, Hungary
2023 - present	Editor in Chief, Advances in Complex Systems, World Scientific

Previous positions (selected)

2020 - 2022	Research Advisor, UNICEF Office of Innovation, NYC, USA
2013 - 2019	Assistant Professor, Ecole Normale Superieure de Lyon, Lyon, France

Selected Awards

2021: Awarded Fellow of the AccelNet-Multinet Program; 2018: Junior Scientific Award of the Complex System Society

Selected Third Party Funding, Grants and Projects

2019 - 2023	ANR (France), DataRedux, € 780.000, PI
2019 - 2023	H2020-ESRAI, SoBigData++, € 318.000 of € 10M, PI (National node)
2024 - 2028	WWTF, MOMA, € 650.000, Co-PI

Supervision: 9 Postdocs, 11 PhDs, 14 MSc research students, 3 BSc students, 3 visiting PhDs

Selected Publications

L. Alessandretti, M. Karsai, L. Gauvin, User-based representation of time-resolved multimodal public transportation networks. R. Soc. Open Sci. 3, 160156 (2016).

JL Abitbol, M Karsai, Interpretable socioeconomic status inference from aerial imagery through urban patterns. Nature Machine Intelligence 2 (11), 684-692 (2020)

RM Hilman, G Iñiguez, M Karsai, Socioeconomic biases in urban mixing patterns of US metropolitan areas. EPJ data science 11 (1), 32 (2022)

L Espín-Noboa, J Kertész, M Karsai, Interpreting wealth distribution via poverty map inference using multimodal data. WWW'23 ACM The Web Conference, Austin, April 30-May 4 (2023).

L Napoli, V Sekara, M García-Herranz, M Karsai, Socioeconomic reorganization of communication and mobility networks in response to external shocks. Proceedings of the National Academy of Sciences (PNAS) 120 (50), e2305285120 (2023)

Staff members

Veronika Csapó, Research Grants Coordinator; *Lisette Espin-Noboa*, Postdoctoral Research Fellow; *Timur Naushirvanov*, PhD Student

Project Partner 3 - César Hidalgo, Toulouse School of Economics

The Toulouse School of Economics (TSE) is a French Grand Establishment well known for its contributions to fundamental and applied economics research. The main task of TSE will be the leadership of WP4 and WP8, to participate in empirical analyses, and to act as national contact point for France. TSE has a robust track record in leading and participating in European Research Projects.

César A Hidalgo joined TSE in 2023 and was promoted to full professor in 2024. Hidalgo has over a decade of research experience on urban topics. He pioneered the use of crowdsourcing and machine learning methods to map urban perception and physical urban change and the use of network methods to understand the amenity composition of neighbourhoods. Professor Hidalgo now leads the Center for Collective Learning (CCL), a multidisciplinary and international research laboratory supported by several European and national awards (ANR). Portfolio: ORCID; Google Scholar.

Education

2008 PhD, Physics, University of Notre Dame, Notre Dame, Indiana, USA
2003 Bachelor, Physics, Universidad Católica de Chile, Santiago, Chile

Current positions

2024 – Professor, Toulouse School of Economics, Toulouse, France
2023 – Head of Research, CCL, CIAS, Corvinus University
2019 – Honorary Professor, Alliance Manchester Business School, University of Manchester, UK

Previous positions (selected)

2019 – 2023 ANITI Chair, Université de Toulouse, France
2019 – 2022 Visiting Professor, SEAS, Harvard University
2015 – 2019 Associate Professor, Media Arts and Sciences, SA&P, MIT, USA
2010 – 2015 Assistant Professor, Media Arts and Sciences, SA&P, MIT, USA
2008 – 2010 Research Fellow, CID, Harvard University, USA

Selected Awards

2018 Lagrange Prize – ISI Foundation, Turin, Italy
2018 Webby Award for Best Use of Machine Learning – Street Change
2018 Webby Award for Best Government and Civil Innovation - Data Africa
2017 Webby Award for Best Government and Civil Innovation - Data USA
2011 Bicentennial Medal of Science – Congress of Chile

Supervision: 7 PhDs, 14 Postdocs, 1 visiting Postdoc, 18 Master students

Selected publications

Hidalgo, César A., et al. "The product space conditions the development of nations." Science 317.5837 (2007): 482-487.
Hidalgo, César A., and Ricardo Hausmann. "The building blocks of economic complexity." Proceedings of the national academy of sciences 106.26 (2009): 10570-10575.
Naik, Nikhil, Scott Duke Kominers, Ramesh Raskar, Edward L. Glaeser, and César A. Hidalgo. "Computer vision uncovers predictors of physical urban change." Proceedings of the National Academy of Sciences 114, no. 29 (2017): 7571-7576.
Jara-Figueroa, Cristian, Bogang Jun, Edward L. Glaeser, and Cesar A. Hidalgo. "The role of industry-specific, occupation-specific, and location-specific knowledge in the growth and survival of new firms." Proceedings of the National Academy of Sciences 115, no. 50 (2018): 12646-12653.
Hidalgo, César A. "Economic complexity theory and applications." Nature Reviews Physics 3, no. 2 (2021): 92-113.

Staff members

Celine Claustre, Head of Financial Control and Grants; *Jairo Gudino*, Research Assistant

Project Partner 4 - Laura Alessandretti, Technical University of Denmark

The Technical University of Denmark (DTU) is an engineering and technology university known for its leading role in technical and natural sciences. DTU consistently ranks among the top technical universities in Europe. The main task of DTU will be the leadership of WP5, participation in further modelling, and acting as a national contact point for Denmark. DTU has an extensive history of leadership and participation in European research projects, particularly through frameworks like Horizon 2020 and the European Research Council.

Laura Alessandretti is Associate Professor at the Technical University of Denmark, Section for Cognitive Systems. Her work focuses on the study Human Behaviour using large-scale data and methods form Network Science, Complex Systems and Computer Science. Her research led to discoveries on fundamental aspects of Human Mobility, such as the trade-off between exploration and exploitation in visiting patterns, the interplay between social and mobility behaviours, and the effects of cognitive constraints on movements. Her works also focuses on topics of societal interest, including gender-gaps in mobility, urban segregation, and the spread of epidemics. Portfolio: ORCID; Google Scholar.

Education

2015 – 2018 PhD in Applied Mathematics; City, University of London; UK
2012 – 2014 MSc in Physics; Ecole Normale Superieure de Lyon; France
2009 – 2012 BSc in Physics; University of Turin; Italy

Current position

Since Feb 2024 Associate Professor, DTU Compute, Sec. Cognitive Systems, København, DK

Previous positions

2020 – 2024 Tenure Track Assistant Professor in Modelling Human Dynamics
2018 – 2020 Postdoctoral Fellow, Copenhagen Center for Social Data Science, København, DK

Selected award

2023 Danish Young Academy's Research Environment of the Year, Det Unge Akademi
2022 Best Italian Researcher in Denmark, The Italian Embassy
2021 Tietgenprisen by the Danish Society of Education and Business (DSEB)

Supervision of graduate students and postdoctoral fellows

Since 2022: 2 PhDs, 2 Postdoctoral fellows

Selected publications

Sapienza, A., Lítlá, M., Lehmann, S., and Alessandretti, L. (2023). Exposure to urban and rural contexts shapes smartphone usage behavior. PNAS nexus, 2(11), pgad357.
Alessandretti, L., Natera Orozco, L. G., Saberi, M., Szell, M., and Battiston, F. (2023). Multimodal urban mobility and multilayer transport networks. Environment and Planning B: Urban Analytics and City Science, 50(8), 2038-2070.
Pappalardo, L., Manley E., Sekara V., and Alessandretti, L. (2023). Future Directions in Human Mobility Science. Nature Computational Science, 3.7, 588-600.
Alessandretti, L., Natera Orozco, L. G., Saberi, M., Szell, M., and Battiston, F. (2022). Multimodal urban mobility and multilayer transport networks. Environment and Planning B: Urban Analytics and City Science, 23998083221108190.
Edsberg Møllgaard, P., Lehmann, S., and Alessandretti, L. (2022). Understanding components of mobility during the COVID-19 pandemic. Philosophical Transactions of the Royal Society A, 380(2214), 20210118.
Alessandretti, L., Aslak, U., and Lehmann, S. (2020). The scales of human mobility. Nature, 587(7834), 402-407.
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Driving Urban Transitions

Project Partner 5 - Riccardo Di Clemente, ISI Foundation

The ISI Foundation (ISI) is a research organization based in Turin, Italy, renowned for its interdisciplinary research in complex systems, network theory, and data science. The main task of ISI will be the leadership of WP3, participation in further empirical work, and acting as the national contact point for Italy. ISI has a strong record of leadership and participation in European research projects, leveraging its expertise in complex systems and computational sciences.

Riccardo Di Clemente leads the multidisciplinary research group at Complex Connection Lab at ISI Foundation and Northeastern University London where he is an Associate Professor. His team innovates mathematical frameworks for analysing complex social connections in cities. Utilizing network theory, computational social science, and machine learning, they explore digital footprints to investigate the digital footprints left by individuals in their daily urban routines to capture the dynamics of cities' segregation. Portfolio: ORCID; Google Scholar.

Education

2014 Ph.D. & Doctor Europæus in Economics, Market and Institution, IMT School for Advanced Studies Lucca with distinctions, visiting Institution Institute for New Economic Thinking at Oxford.
2010 M.Sc. In Theoretical Statistical Physics, University of Rome "Sapienza".
2006 B.Sc. in Physics, University of Rome "Sapienza".

Current Positions

2023–Now Affiliate Researcher, ISI Foundation, Turin, ITA.
2023–Now Associate Professor, Northeastern University London, London, GBR.
2021–Now Visiting Professor, Sony Computer Science Lab, Paris, FRA.

Previous Positions *(selected)*

2020–2023 Lecturer in Data Science, Computer Science Dep. University of Exeter, GBR.
2018–2020 Newton International Fellow, Centre for Advanced Spatial Analysis, UCL, GBR.
2019–2019 Short Term Consultant, Urban Disaster Risk Management, World Bank, USA.

Selected Fellowships and Awards

2021–2023 Alan Turing Fellow, The Alan Turing Institute GBR.
2018–2020 Newton International Fellow, Royal Society, GBR (Grant NF170505).
2019 The World Bank / GFDRR (Contract N. 7194851).
2015 Gates Foundation (Grant N. OPP1141325) & U.N. (Grant N. UNF-15-738).

Supervision Of Postdocs, Ph.D. Students, Graduate Students

2 Postdocs, 7 PhDs, 21 MSc research students, 27 BSc research students, 2 visiting PhDs

Selected Publications (selected relevant to the project within three years)

Ubaldi E., Yabe T., Jones N., Khan M-F., Feliciotti A., Di Clemente R., Ukkusuri S-V., Strano E. Mobilkit: A Python Toolkit for Urban Resilience and Disaster Risk Management Analytics, Journal of Open Source Software, DOI:10.21105/joss.05201 (2024).
Santana C., Botta F., Barbosa H., Privitera F., Menezes R., Di Clemente R., COVID-19 is linked to changes in the time-space dimension of human mobility, Nature Human Behaviour, DOI: 10.1038/s41562-023-01660-3 (2023).
Collins T.R., Di Clemente R., Gutierrez-Roig M., Botta, F. Spatiotemporal gender differences in urban vibrancy, Environment and Planning B: Urban Analytics and City Science, DOI: 10.1177/23998083231209073 (2023).
Toth G., Wachs, J., Di Clemente R., Jakobi A., Sagvari B., Kertesz J. Lengyel B., Inequality is rising where social network segregation interacts with urban topology, Nature Communications DOI: 10.1038/s41467-021-21465-0 (2021)
Di Clemente R., Strano E., Batty M., Urbanization and Economic Complexity, Scientific Reports DOI: 10.1038/s41598-021-83238-5 (2021)

Staff members
Giuliana Antrilli, Communications Department Coordinator

Project Partner 6 - László Lőrincz, Corvinus University of Budapest

Corvinus University of Budapest (CUB) is one of the most prominent research universities in Hungary. The core research and teaching activities have been traditionally performed in the areas of Business Management, Economics, Social Sciences and International Relations, while new specialisations are developing dynamically focusing on current issues, such as Sustainable Development, Data Science and Business Informatics. The main task of CUB will be the leadership of WP7, and involvement in other WPs. CUB has wide-range experience in participation and leadership of H2020 projects. Related research capacities have accumulated in the Networks, Technology, & Innovation (NETI) Lab, a group established in 2020 within Corvinus Institute for Advanced Studies and the Institute of Data Analytics and Information Systems.

László Lőrincz is an Associate Professor at Corvinus University, and a Research Fellow at HUN-REN Centre for Economic- and Regional Studies. He joined the Research Centre for Economic and Regional Studies in 2013, after working in the government, and in the private sector. His current research includes urban mobility, and network effects in labour mobility and in migration. His work was published in leading field journals like Social Networks, Journal of Economic Geography and Network Science. Portfolio: Google Scholar.

Higher Education

2009 Ph.D in Sociolology, Corvinus University of Budapest
2002 MSc in Economics and Sociology, Budapest University of Economics and Public Administration

Current Positions

2023–Now Associate professor, Institute of Data Anaytics and Information Systems, Corvinus University of Budapest
2020–Now Senior Research Fellow at NETI Lab, Corvinus Institute for Advanced Studies
2013–Now (Senior) Research Fellow, HUN-REN Centre for Economic and Regional Studies

Previous Positions (Selected)

2016–2020 Assistant Professor, Centre for Labor Economics, Corvinus University of Budapest, HUN
2012–2013 Head of Unit, ECOSTAT Government Centre for Economic Impact Assessment, Budapest, HUN

Selected Fellowships and Awards

2022 Bolyai+ Scholarship (Hungarian Research, Development and Innovation Office)
2021 Janos Bolyai Scholarship (Hungarian Academy of Sciences)

Selected Publications (selected relevant to the project within three years)

1. Juhász, S., Pintér, G., Kovács, Á. J., Borza, E., Mónus, G., Lőrincz, L., & Lengyel, B. (2023). Amenity complexity and urban locations of socio-economic mixing. *EPJ Data Science*, *12*(1), 34.
2. Ilyés, V., Boza, I., Lőrincz, L., & Eriksson, R. H. (2023). How to enter high-opportunity places? The role of social contacts for residential mobility. *Journal of Economic Geography*, *23*(2), 371-395.
3. Lőrincz, L., & Németh, B. (2022). How social capital is related to migration between communities?. *European Journal of Population*, *38*(5), 1119-1143.

Staff members
Krisztina Hollósi, Research Funding Advisor; *Gergely Mónus*, PhD Student; *Gergő Pintér*, Postdoctoral Researcher; *Géza Salamin*, Associate Professor, member of the European Council of Spatial Planners

Project Partner 7 - Budapesti Közlekedési Központ

Co-operation Partner, Country: Hungary

Contact person: Tamás Halmos, Head of Strategic Knowledge Centre and R&D

BKK Centre for Budapest Transport is the transport organiser of Budapest, it was established in 2010. BKK as a public service provider is fully owned by the Municipality of Budapest. BKK is responsible not only for public transport in Budapest, but also for the whole travel chain (micromobility, curbside management, city logistics etc.) in the city. Their experts take part in the preparation and realization of all major transport-related changes and developments. While doing planning work, probably one of the most important aspects is to create as liveable a city as possible for everyone by coordinating transport solutions of different purposes and speeds. BKK is determined to be an active participant of each international transport organizing network and also of such transport development-related projects, promoting exchange of ideas and innovation, which BKK deems useful and beneficial for the City of Budapest.

Project Partner 8 - Magyar Telekom

Co-operation Partner, Hungary

Contact person: Anita Hodosán-Hartai, Wholesale Partner and Product Expert

Magyar Telekom is not only the biggest telecommunication provider in Hungary but it is also a determining stakeholder, investor and employer of the Hungarian economy. Magyar Telekom is dedicated to help Hungary in digitalization and trying to innovate in as many territories as possible. It is not a coincidence that the competitiveness strategy of Hungary defined the Infocommunication sector as a disruptive one and a focus sector. This is due to the high technology such as cloud, IoT or MI. However, data-driven decision making and geolocational data and geolocational analysis are also cutting-edge solutions. Magyar Telekom has wide experiences in B2B projects using locational data to help the companies to optimize their costs, support partners' marketing campaigns, help them to find the next best spot for their franchise or help the tourism agencies with location data. The company also runs many projects to increase its engagement with the society as well. For example, Magyar Telekom is actively participating in the rationalization of urban transport, the shaping of the shopping mix of plazas, the utilization and development of parks, and the construction of parking lots.

Project Partner 9 - Budapest Főváros Önkormányzata

Co-operation Partner, Hungary

Contact person: Dóra Anna Kókai, Head of Project Management Unit

The Municipality of Budapest is responsible for the services that fundamentally determine the functioning of the city, and these must be developed in the coming years, considering changing consumer habits and social needs, using new technologies, so that they operate in a financially sustainable way, reduce environmental damage and, finally, increase the satisfaction of the city's residents and customers. Therefore, to achieve these goals, the main task of the Municipality of Budapest is to promote an environmentally friendly, resource-efficient green transition and digitalisation, making the best use of the opportunities offered by technological progress and innovation, both in the economy and in the renewal of urban services and the urban environment. These orientations are closely aligned with the European Union's objectives to undertake the most important structural changes over the next decade to contribute to global climate goals and improve Europe's competitiveness.

The Municipality of Budapest is a beneficiary of several major EU-funded infrastructure development projects and a partner in many other soft projects.

Project Partner 10 - Urban Innovation Vienna

Co-operation Partner, Austria

Contact person: Gerald Franz, Teamlead

As the climate and innovation agency of the City of Vienna, Urban Innovation Vienna works on the successful development of cities and makes metropolises 'fit for the future'. In doing so, the company can point to extensive expertise in our core areas of mobility, energy, digitalisation, urban and district development. They act as an impulse generator and innovation driver and work with our multidisciplinary team in a professional and committed manner on the transformation of cities. They support a wide range of actors, from administration and politics to business and science, promote multi-stakeholder dialogue and incorporate participation and co-creation as a vital element in the development of effective and sustainable solutions. To accelerate the change towards a more sustainable mobility system and climate neutrality, the first "Mobility Policy Innovation Lab" (in short: “Policy Lab”) was established in September 2022. The Lab not only develops forward-looking mobility measures, but also looks at which governance instruments can be used to implement them. To this end, it deals with specific challenges and tasks that are brought to the Lab by the federal government, federal states, regions, and cities. UIV Urban Innovation Vienna is the Policy Lab’s lead organisation. Urban Innovation Vienna coordinates all project activities and sets priorities in terms of addressed topics and issues.

Project Partner 11 - Toulouse Metropole

Co-operation Partner, France

Contact person: Audeline Rauna, Head of External Financing

Toulouse Métropole, as a European metropolis, a major international reference in the aeronautics and space industries and the capital of the Occitanie region boasts a dynamic economy and a quality of life that attract thousands of new arrivals every year. Drawing on a particularly well-developed and dynamic industrial, university and entrepreneurial ecosystem, Toulouse has embarked on a remarkable economic diversification into many of the sectors of the future: digital, health and life sciences, robotics, transport of the future and the Internet of Things, etc. Innovation and Research & Development are at the heart of Toulouse's identity. There are more than 22,000 researchers, 400 research units and some 120,000 students in this region of excellence. Toulouse Métropole fosters 5 innovation hubs dedicated to specific sectors: aerospace, mobility, health, food, digital. They gather in one same spot some businesses, mainly startups, but also incubators / accelerators and sometimes facilitators. For example:

1) *Grand Matabiau Digital Hub* is the gathering, in a brand-new building at the heart of Toulouse city centre, of actors in the field of digital businesses, AI and cybersecurity. It will host innovation actors like digital services businesses, engineering counselling societies, deeptech startups, digital clusters, or researchers on AI. 2) *Montaudran Aerospace Hub* and *B612 Innovation Centre* gather entities dedicated to aeronautics, space, and embedded systems. Its symbol is the Toulouse Métropole owned B612 Building, an Innovation centre hosting more than 10 actors in the field of aerospace, including the famous competitiveness pole Aerospace Valley. 3) *Francazal Mobility Hub* is the rehab of a former military base into an industrial and technological hub dedicated to innovative mobility (eg. autonomous shuttle, electric plane) and low carbon technologies (hydrogen “technocampus”). It fosters the development of such an ecosystem while using environmentally friendly methods and hosts a dozen actors, from startups to bigger businesses.

Toulouse Métropole pilots a large number of European projects and has extensive experience of European programmes: LIFE, Horizon Europe, MIE, CEF etc. It is also the main partner of the European Institute of Innovation and Technology (EIT Urban Mobility).

Project Partner 12 - 5T SRL

Co-operation Partner, Italy

Contact person: Giandomenico Gagliardi, International R&I Project Manager

The company 5t specializes in intelligent and sustainable mobility governance, leveraging its extensive experience and competence since 1992. They offer tools and services to manage territorial mobility, carefully selecting top technological solutions. 5t is responsible for key mobility centres in Torino and the Piemonte Region, ensuring daily operations and providing data and analyses to support policymaking. They deliver up-to-date information and real-time infomobility services to citizens and visitors, leveraging open-source technology. Notably, they designed and manage the first regional ticketing system, continually enhancing accessibility. 5t prioritizes sustainable and sharing mobility, leading strategic projects within the Mobility as a Service (MaaS) and Smart Road sectors. They are recognized nationally and across Europe for their expertise in transport standardization, actively engaging in innovative projects and publishing cutting-edge research. Additionally, 5t participates in a global network dedicated to collaboration and the exchange of new mobility ideas.

Project Partner 13 - The City of Copenhagen

Co-operation Partner, Denmark

Contact person: Anna Garrett, Team Manager, Mobility analysis and planning

Copenhagen is at the forefront of urban sustainability, particularly highlighted by its emphasis on active mobility, public transportation, and sustainable urban development. Renowned as one of the world's most bicycle-friendly cities, Copenhagen has invested heavily in a vast network of safe and accessible bicycle lanes, alongside pedestrian-friendly zones, encouraging residents to opt for biking or walking over motorized transport. Complementing this, the city boasts an efficient public transportation system, recently enhanced by the expansion of its metro with the City Circle Line, facilitating seamless and eco-friendly commutes across diverse neighbourhoods. These initiatives are part of a broader sustainable urban development strategy aimed at making Copenhagen carbon-neutral by 2025. This includes incorporating energy-efficient technologies and green spaces into urban planning and development, ensuring all new constructions align with environmental sustainability goals, and revitalizing former industrial zones into mixed-use areas to maintain a vibrant urban core that supports a sustainable lifestyle.

Project Partner 14 - Toretei SRL

Co-operation Partner, Italy

Contact person: Elizaveta Gazeeva, Co-founder, UX/UI designer

UrbanistAI is a Generative AI platform designed for participatory planning and co-design in urban settings. UrbanistAI represents a methodological shift towards a more inclusive approach in urban planning, combining community involvement with technological innovation to reshape urban spaces. UrbanistAI tool was created through a collaboration between Toretei and SPIN Unit, bringing together advanced AI development and urban planning expertise. Toretei is an Italian high-tech company, which develops AI-powered design software solutions, and offers consulting services to help other companies integrate generative AI technologies into their design processes. The experience of Toretei spans working with city administrations, urban planning consultancies, universities, and international organizations.

7.5. Consortium experience and complementary with other projects of the partners

The COLINE consortium has collected significant experience in the topic of the proposal and Co-applicants have published in globally leading journals in the field of urban mobility, segregation, urban perception etc. In the last three years, the Main Applicant and Project Partner 2 have participated in projects that focused on segregation and mixing in urban environments.

The OTKA K 138970 funded by NKFIH and led by the Main Applicant is still ongoing until August 2025 and focuses on urban mobility and socio-economic mixing. Empirical knowledge has already accumulated in this project about the role of amenities in social mixing.

The Main Applicant has led the LP2017-6 Lendület project funded by the Hungarian Academy of Sciences. In this project, social network segregation was found to be correlated with characteristics of the built environment and was claimed to be an important factor of inequality dynamics. Further, the urban poor were found to have spatially concentrated social networks but can establish links with other socio-economic groups in case they commute to a workplace that is distant from their home location.

Project Partner 2 has been participating in a H2020 project “H2020-871042”, in which the aim was to create high resolution maps that can capture spatial income segregation in data poor environments. Further, data on telecommunication interactions were used to investigate the dynamics of network segregation during the COVID-19 pandemic.

Although the above projects are related to the recent proposal, neither of the objectives of COLINE were addressed before by the consortium.

Table 7.1: Existing results and deliverables obtained from publicly funded projects which provide the basis of or feed into the proposed project

Funding provider	Project number	Title	Description of results already obtained and relevant deliverables (verifiable results / products of R&D work) in terms of the basis for / differentiation from the proposed project	Location and type of documentation (e.g. link to homepage, publication, conference proceedings, interim report, final report, …)
NKFIH	OTKA K 138970	Mobility, social- and economic networks, and inequalities in cities over the pandemic	Using GPS mobility data, we demonstrate that neighbourhoods that are centrally located and have diverse non-ubiquitous amenity portfolios facilitate socio-economic mixing. Major roads are not only channels but also barriers of mobility, especially for those who live close to the city centre.	https://epjds.epj.org/articles/epjdata/abs/2023/01/13688_2023_Article_413/13688_2023_Article_413.html https://arxiv.org/abs/2312.11343
H2020-ESRAI	H2020-871042	Integrated Infrastructure for Social Mining and Big Data Analytics	Using multiple source open data we infer high-resolution socioeconomic maps to understand the segregation dynamics characterizing re-organised mobility and social networks during external shocks.	https://www.pnas.org/doi/abs/10.1073/pnas.2305285120 https://dl.acm.org/doi/10.1145/3543507.3583862
MTA Lendület	LP2017-6	Agglomeration and social networks	Using social media data, we find that income inequalities grow in those urban areas where social networks are fragmented due to the fragmented structure of the built environment. We find that commuting to work increases the ability of citizens living in relatively poor neighbourhoods to establish social relationships with other socio-economic groups.	https://www.nature.com/articles/s41598-021-00416-1 https://www.nature.com/articles/s41467-021-21465-0

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